Accounting for the End of Orbit

An Orbital Retirement, Decommissioning and Debris Obligation Model under IFRS and US GAAP

July 2026

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Author's Note

This academic working paper is intended for technical discussion among chief financial officers, controllers, auditors, satellite and launch operators, telecommunications companies, insurers, orbital-servicing providers, valuation specialists, standard setters, securities regulators, licensing authorities, lenders, and capital-market participants. The Orbital Retirement, Decommissioning and Debris Obligation Model (ORDOM) proposed herein is not an official interpretation of IFRS Accounting Standards, US GAAP, SEC rules, communications regulation, space law, environmental law, insurance law, or aerospace-engineering requirements. It is a falsifiable accounting framework designed to translate end-of-mission architecture into a disciplined analysis of present obligations, unit of obligation, recognition, scenario-based measurement, subsequent remeasurement, asset-liability interaction, settlement, internal controls, and disclosure.

ORDOM is the fourth layer of an integrated space-accounting architecture. The Mission-Cycle Depreciation Model (MCDM) measures the consumption of reusable launch assets. The Reusable Launch Cost Allocation Model (RLCAM) assigns launch-system cost to external contracts, internally constructed orbital assets, research activity, or period expense. The Network-Utility Impairment Model (NUIM) tests whether the resulting orbital and network carrying amounts remain recoverable. ORDOM addresses the next question: when placing and operating an object in orbit creates legal, contractual, or constructive retirement duties, how should the entity identify, measure, update, settle, and disclose the related obligation?

The public-company and regulatory examples distinguish strictly between observed data and illustrative assumptions. Iridium Communications, Spire Global, Sirius XM, and other operators do not publicly disclose the complete license conditions, legal opinions, asset-level cost registers, reserved propellant, disposal probability models, active-removal contracts, or discount-rate layers required to execute ORDOM numerically. Whenever the paper presents a calculation beyond disclosed amounts, the assumptions are explicitly identified as hypothetical and are not represented as the accounting records, policies, estimates, or actual economics of any named entity. The paper does not infer noncompliance by any company (Iridium Communications, 2015, 2026; Sirius XM Holdings, 2026; Spire Global, 2026).

Abstract

Orbital infrastructure creates retirement obligations that conventional decommissioning models do not map cleanly. A satellite can be physically unreachable, economically embedded in a constellation, licensed under an operator-specific disposal plan, capable of natural decay, dependent on reserved propellant and ground control for self-disposal, or subject to a fallback active-removal strategy. The same end-of-mission architecture can therefore contain a recognized asset retirement obligation, an unrecognized operating activity, a contingent liability, an impairment or derecognition event, and a sustainability commitment. IAS 16, IAS 37 and IFRIC 1 under IFRS, and ASC 410-20 under US GAAP, provide authoritative principles for retirement obligations but do not provide a sector-specific method for identifying the obligating event, unit of obligation, measurement scenarios, telemetry inputs, and disclosure perimeter for space assets.

This paper develops the Orbital Retirement, Decommissioning and Debris Obligation Model (ORDOM), an engineering-to-accounting framework for recognizing and measuring obligations associated with controlled or uncontrolled reentry, orbit lowering, transfer to a disposal orbit, passivation, post-mission tracking, collision-risk mitigation, contracted servicing, active debris removal, and final asset derecognition. ORDOM separates the physical asset, the economic impairment unit, and the legal or constructive obligation unit; imposes a present-obligation gate before measurement; builds a scenario-complete cost stack; uses telemetry to update probabilities without allowing engineering precision to replace accounting judgment; and preserves the different recognition and measurement mechanics of IFRS and US GAAP.

Public evidence demonstrates both the relevance and the accounting boundary. The FCC's 2022 order shortened the post-mission disposal benchmark to five years for covered non-geostationary stations using uncontrolled atmospheric reentry and requires licensing disclosures about disposal plans and fuel reserved for disposal. ESA's Zero Debris approach uses a disposal-success threshold above 90%, continuous monitoring of critical disposal functions, and removal interfaces for missions that may require active removal. Iridium's 2025 Form 10-K describes a 66-satellite operational constellation whose second-generation satellites remain subject to a 25-year de-orbit standard under its current FCC authorization; historical Iridium filings separately reported an asset retirement obligation linked to contractual mass-deorbit rights. Spire reported US$9.1 million of losses on decommissioned satellites and other asset write-offs in 2025, illustrating that asset derecognition and liability measurement answer different questions (FCC, 2022; ESA, 2023, 2025; Iridium Communications, 2015, 2026; Spire Global, 2026).

Methodologically, ORDOM is framed as a Design-Science Research (DSR) contribution in financial accounting and provides a normative blueprint for standard setters, regulators, auditors, and preparers rather than a claim of completed empirical validation. It defines recognition boundaries, obligation units, scenario trees, cost categories, discounting controls, remeasurement rules, portfolio aggregation limits, anti-management-override procedures, and a three-layer disclosure architecture. Its central proposition is that economic continuity at the network level does not extinguish object-specific retirement obligations, while societal concern about orbital debris does not by itself create a recognized liability without an enforceable or constructive present obligation under the applicable reporting framework.

1. Introduction

The financial-accounting lifecycle of a space asset can be decomposed into four analytically distinct layers. MCDM measures how a reusable launch system is consumed by missions. RLCAM assigns the resulting launch-system cost to external contracts, orbital assets, research activity, inventory, or period expense. NUIM determines the economic unit at which recognized orbital assets are tested for recoverability. ORDOM addresses the separate question of obligations created by the end of an orbital asset's useful or licensed life. This sequence is conceptual rather than an implementation prerequisite: ORDOM is a fully autonomous, modular framework that can be applied without MCDM, RLCAM, or NUIM, including by entities using straight-line depreciation under IAS 16 or ASC 360 and conventional impairment testing under IAS 36 or ASC 360.

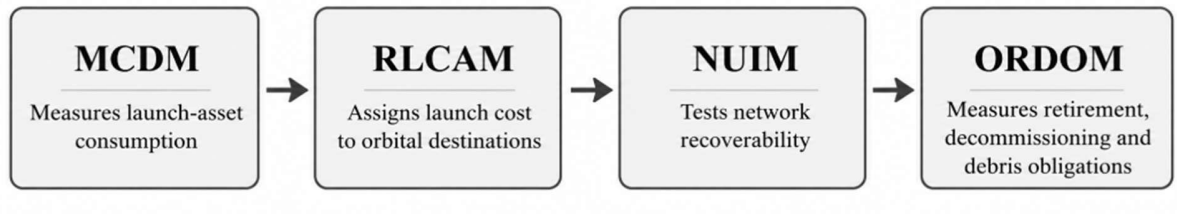


Figure 1. ORDOM as the fourth layer of the integrated space-accounting architecture.

The accounting problem begins with an apparent paradox. A satellite can cease generating economic benefits long before it leaves orbit, yet the cash outflow required to dispose of it may occur years later. A spacecraft can remain technically functional but be retired because a replacement generation provides better capacity. It can be removed from commercial service and retained as an in-orbit spare. It can lose communications and become incapable of self-disposal. It can be designed to decay naturally without a terminal maneuver. It can be transferred to a graveyard orbit. It can require a contracted active-removal mission. Each fact changes the expected cash flows, but none independently determines whether a present obligation exists or how that obligation should be measured.

The problem is becoming more material as orbital architectures scale and regulatory expectations tighten. The FCC's five-year rule applies to covered non-geostationary stations ending their missions in, or passing through, low-Earth orbit and using uncontrolled atmospheric reentry. The rule shortened the prior 25-year benchmark and preserved authorization-specific transition provisions. ESA's Zero Debris approach similarly emphasizes shorter orbital clearance, verified disposal-success probabilities, continuous monitoring of disposal-critical functions, passivation, collision avoidance, and interfaces that enable active removal if self-disposal fails. NASA requires debris assessment and end-of-mission planning for missions within its authority. These developments convert disposal from a remote engineering afterthought into a mission-design, licensing, governance, and cash-flow issue (FCC, 2022; ESA, 2023, 2025; NASA, 2021, 2024).

Financial reporting cannot respond by recording the social cost of every object placed in orbit. A liability requires an obligation under the applicable framework. Under IFRS, that obligation may be legal or constructive, but it must result from a past event and meet the recognition conditions in IAS 37. Under US GAAP, ASC 410-20 applies to legal obligations associated with retirement of tangible long-lived assets; uncertainty about timing or method does not eliminate a conditional legal obligation if fair value can be reasonably estimated. A voluntary target, a general industry guideline, or management's current intention to behave responsibly is not automatically equivalent to a recognized obligation.

The opposite error is equally serious. An entity can conclude that deorbiting is part of ordinary operations and therefore omit an obligation that has already been incurred through a license, contract, law, or valid constructive commitment. It can use a high expected self-disposal rate to exclude the cost of fallback removal. It can aggregate a large constellation and assume that failures offset one another even when each space object has an independent legal disposal requirement. It can write off the satellite but leave the retirement liability unchanged, or reduce the liability because the asset has already been impaired. It can classify disposal expenditure as future operating cost even though the obligating event occurred at launch or during operation.

ORDOM is designed to prevent these errors without creating accounting rules outside IFRS or US GAAP. The model begins with legal and constructive-obligation analysis, not a cost estimate. It identifies the object or portfolio to which the obligation attaches. It maps the disposal method and fallback paths. It develops probability-weighted cost scenarios using engineering, contractual, market, and regulatory evidence. It then applies the recognition, discounting, capitalization, remeasurement, settlement, and disclosure requirements of the applicable reporting framework.

Methodologically, the paper follows a Design-Science Research logic: it diagnoses a recurring reporting problem, designs an auditable accounting artifact, specifies its decision gates and control architecture, and states boundary conditions and testable propositions. ORDOM is therefore offered as a normative blueprint for the IASB, FASB, SEC, FCC, auditing firms, and preparers, not as a descriptive claim that current aerospace reporting practice is uniform.

Core thesis

Economic continuity at the constellation level does not extinguish object-specific retirement obligations. Conversely, the existence of orbital-debris risk does not create a financial-statement liability unless the entity has a present obligation under the applicable accounting framework.

2. Research Question and Contributions

This paper addresses the following research question:

Research question

How should satellite and launch operators identify, measure, update, settle, and disclose orbital retirement, decommissioning, and debris-related obligations under IFRS and US GAAP when legal duties, constructive commitments, disposal technology, mission telemetry, and network economics do not align at the same unit or time?

The paper makes seven contributions.

1. It separates the asset unit, depreciation unit, impairment unit, and obligation unit. A satellite can be recorded and depreciated individually, tested for recoverability as part of a network under NUIM, and simultaneously carry an object-specific retirement obligation.
2. It introduces a present-obligation gate that prevents expected retirement cost from being measured before the entity establishes the legal, contractual, or constructive source of the duty and the relevant past event.
3. It develops a scenario-complete measurement architecture covering self-disposal, partial failure, natural decay, graveyard-orbit transfer, controlled reentry, passivation, extended tracking, contracted servicing, and active removal without treating every externality as the entity's liability.
4. It translates telemetry into accounting evidence. Propellant, propulsion health, attitude control, command-and-control capability, orbital lifetime, conjunction exposure, and removal-interface readiness update probabilities and timing, but do not replace the standards' recognition criteria.
5. It establishes anti-shielding and anti-double-counting controls. Portfolio aggregation cannot erase object-level duties; failure risk cannot be reflected simultaneously in scenario probabilities, cash-flow contingencies, and discount rates; and a satellite write-off cannot automatically extinguish the related liability.
6. It preserves the different IFRS and US GAAP mechanics for constructive obligations, probability thresholds, measurement objectives, discount rates, accretion or unwinding, and subsequent revisions.
7. It proposes a three-layer disclosure architecture connecting the financial-statement note, critical judgments and estimates, and operational sustainability metrics while protecting mission-sensitive telemetry and commercially sensitive servicing terms.

Table 1. ORDOM contributions and boundary conditions.

Contribution area	ORDOM contribution	Boundary condition
Recognition	Uses an obligation-source and past-event gate before estimating cost.	Does not recognize a provision solely because disposal is environmentally desirable or operationally planned.
Unit of obligation	Links the duty to the spacecraft, upper stage, launch batch, license, or homogeneous portfolio that bears it.	Does not default automatically to the entire constellation or consolidated company.
Measurement	Uses complete, probability-weighted disposal and fallback scenarios with a controlled cost stack.	Does not include societal externalities, penalties, or collision damages unless the entity has the relevant present obligation.
Telemetry	Updates settlement probability, timing, and method from controlled engineering evidence.	Does not convert telemetry precision into legal certainty or override accounting recognition criteria.
Asset interaction	Coordinates capitalized retirement cost, depreciation, impairment, derecognition, and liability settlement.	Does not eliminate a liability merely because the related satellite has been written off.
IFRS-US GAAP bridge	Applies a common operational fact base through separate accounting measurement engines.	Does not blend IAS 37 best estimate with ASC 410-20 fair value.
Disclosure	Reconciles recognized amounts, obligation scope, timing, assumptions, and sensitivity.	Does not require disclosure of security-sensitive orbital or propulsion data.

3. Regulatory and Conceptual Architecture

3.1 IAS 16: the asset retirement cost

IAS 16 includes in the cost of property, plant and equipment an initial estimate of the costs of dismantling and removing the item and restoring the site on which it is located when the entity incurs the obligation. For orbital assets, the application is not mechanical. Outer space is not a conventional owned or leased terrestrial site, and many disposal actions do not restore a physical location to its former condition. The relevant analysis is whether the recognized obligation is associated with retirement of the tangible asset and whether the related expenditure is necessary because the entity acquired, constructed, launched, or operated that asset.

When the obligation exists at acquisition, construction, launch, or placement into service, the corresponding amount can qualify as part of the asset's cost under IAS 16, subject to IAS 37 measurement. The debit should be attributed to the asset or component whose operation created the duty. A fleet-level provision should not be capitalized indiscriminately across unrelated spacecraft. If the obligation is incurred later because of a new law, license modification, contract, or constructive commitment, IFRIC 1 determines how the change interacts with the carrying amount of the asset (IFRS Foundation, IAS 16; IFRIC 1).

3.2 IAS 37: present obligation, probability, and best estimate

IAS 37 requires a provision when an entity has a present legal or constructive obligation as a result of a past event, an outflow of resources embodying economic benefits is probable, and a reliable estimate can be made. The obligating event must leave the entity with no realistic alternative to settlement. For an orbital mission, the candidate past event could be launch, orbital insertion, operation under a license, completion of a specified activity, public commitment that creates a valid expectation, or failure to complete a mandated disposal maneuver. The event depends on the source and terms of the obligation; it is not selected merely to accelerate or defer recognition (IFRS Foundation, IAS 37).

A legal obligation may arise from legislation, regulation, a license, a contract, an indemnity, or the legal construction of those instruments. A constructive obligation requires an established pattern of past practice, published policy, or sufficiently specific current statement that creates a valid expectation in other parties that the entity will accept and discharge the responsibility. Under IAS 37, voluntary ESG

targets or broad corporate aspirations do not constitute a recognized constructive obligation unless the following three conditions are met cumulatively:

1. ***Specificity:*** The commitment clearly identifies the specific asset, generation, or constellation cohort subject to disposal.
2. ***External Reliance:*** The policy has been formally communicated to regulators, insurers, lenders, or customers, creating a valid expectation that the entity will accept the duty.
3. ***Unavoidability:*** The entity faces severe operational, regulatory, or commercial consequences if it attempts to withdraw from the public commitment, leaving no realistic alternative to settlement.

Failure of any one criterion keeps the statement within governance, risk, or sustainability disclosure rather than the recognized provision. Management intent, board approval, or participation in a voluntary industry pledge is insufficient on its own.

The provision is measured at the best estimate of the expenditure required to settle or transfer the present obligation at the reporting date. Expected value is normally appropriate for a large population of homogeneous objects or a range of outcomes. For a single spacecraft with a limited number of discrete disposal outcomes, the most likely outcome may require adjustment for other possible outcomes. Risks and uncertainties are reflected without deliberate overstatement. Future events, such as lower-cost active-removal services, are included only when there is sufficient objective evidence that they will occur. Gains on expected disposal of assets are excluded from the provision.

3.3 Discounting, risk, and the IAS 37 improvement project

When the effect of the time value of money is material, IAS 37 requires present-value measurement using a pre-tax rate reflecting current market assessments of the time value of money and risks specific to the liability that are not already reflected in cash flows. ORDOM therefore prohibits double-counting: if disposal failure, cost escalation, or execution risk is represented in scenario probabilities or cash-flow amounts, the same risk cannot be added again through an arbitrary risk premium in the discount rate.

The IASB's Provisions—Targeted Improvements project is especially relevant to orbital obligations. The 2024 exposure draft proposed clarifications on recognition, costs included in measurement, and discount rates. In September 2025, the IASB tentatively supported a risk-free rate with no non-performance-risk adjustment. In February 2026, the Board tentatively decided to clarify why asset-decommissioning and environmental-rehabilitation obligations satisfy a transfer condition. The project remained in redeliberation in June 2026, with final amendments identified as the next milestone. ORDOM therefore applies current IAS 37 while separately identifying assumptions that may require transition if the final amendments change measurement or disclosure requirements (IFRS Foundation, 2024, 2025, 2026).

3.4 IFRIC 1: subsequent changes

IFRIC 1 addresses changes in existing decommissioning, restoration, and similar liabilities caused by revisions in the estimated timing or amount of outflows or changes in the discount rate. Under the IAS 16 cost model, an increase is generally added to the related asset and a decrease deducted from it. A decrease exceeding the asset's carrying amount is recognized immediately in profit or loss. An increase requires consideration of whether the adjusted asset is impaired. Once the related asset has reached the end of its useful life, subsequent changes are recognized in profit or loss as they occur because no future benefit remains to absorb a capitalized adjustment (IFRS Foundation, IFRIC 1).

Impairment Ceiling Rule. Under IFRIC 1, increases in an existing liability are generally added to the related asset's carrying amount under the IAS 16 cost model. ORDOM applies an immediate ceiling control: if an upward revision occurs when the underlying spacecraft is fully impaired, derecognized, or has no remaining recoverable amount, the asset cannot absorb the debit. Any adjustment that would leave the asset above its recoverable amount under IAS 36 is recognized immediately in profit or loss as an operational decommissioning expense. Under US GAAP, a related ARC layer is subject to ASC 360: if no future economic benefit remains, the debit is expensed; otherwise the asset or asset group is evaluated under ASC 360's recoverability and impairment-measurement framework, or at fair value less cost to sell when classified as held for sale.

This interaction is critical for failed satellites. Loss of command capability can increase the expected cost of disposal while simultaneously reducing the asset's recoverable amount to zero. The asset write-off does not cancel the obligation; instead, the liability can increase at the same time the remaining asset

balance is impaired or derecognized. ORDOM requires the two analyses to be documented separately and reconciled.

3.5 Reimbursements, insurance, and decommissioning funds

Expected insurance recoveries, government indemnities, supplier guarantees, launch-service credits, and decommissioning-fund rights do not eliminate the gross obligation unless the entity is legally released. Under IAS 37, a reimbursement asset is recognized only when it is virtually certain that reimbursement will be received if the entity settles the obligation, and it cannot exceed the provision. IFRIC 5 applies when entities contribute to a separate fund established to reimburse decommissioning costs. ORDOM therefore records the liability and reimbursement right on separate evidentiary tracks.

3.6 ASC 410-20: legal asset retirement obligations

ASC 410-20 applies to legal obligations associated with the retirement of tangible long-lived assets that result from acquisition, construction, development, or normal operation. A legal obligation can arise from enacted law, statute, ordinance, written or oral contract, or legal construction of a contract. The liability is initially recognized at fair value when incurred if a reasonable fair-value estimate can be made, with an associated asset retirement cost capitalized as part of the long-lived asset.

The US GAAP scope is narrower than IFRS in one important respect: a constructive obligation without a legal basis is not an ASC 410-20 asset retirement obligation. Conversely, a legal obligation whose timing or method is conditional on a future event is not avoided merely because management expects the triggering event not to occur. FIN 47, now codified in ASC 410-20, clarifies that uncertainty about timing or method does not eliminate the unconditional legal duty. Expected present-value techniques can be used to estimate fair value across possible settlement dates, methods, and amounts.

3.7 US GAAP subsequent accounting

After initial recognition, the asset retirement cost is depreciated over the useful life of the related asset and the liability is increased through accretion expense. ASC 410-20 historically uses a credit-adjusted risk-free rate for each liability layer. Upward revisions are discounted using a current rate; downward revisions are associated with the rate applied to the related prior layer. Settlement produces a gain or loss when actual cash outflows differ from the recorded liability. These mechanics differ materially from current-rate remeasurement under IFRS and must remain separate in dual-reporting models.

3.8 Impairment, derecognition, and contingencies

An orbital retirement obligation is not an impairment model. NUIM determines whether the carrying amount of the satellite or network is recoverable. ORDOM determines whether a present obligation exists and how the liability is measured. A decommissioning decision can trigger both analyses, but the accounting entries have different objectives. A satellite can be fully impaired while the liability remains, or it can retain economic value while a retirement obligation is recognized and accreted.

Collision damages, regulatory penalties, third-party claims, or remediation duties created by a specific accident may fall outside the ordinary asset-retirement obligation and require separate analysis under IAS 37, ASC 450, ASC 410-30, insurance guidance, and applicable law. ORDOM includes those amounts only when they are part of the retirement obligation's enforceable settlement scope. It does not convert a general risk of future collision into a provision.

3.9 Space-law and regulatory evidence

International treaties and nonbinding guidelines shape licensing and risk allocation but do not automatically create a corporate accounting liability. The Outer Space Treaty and Liability Convention primarily address state responsibility and launching-state liability. National licenses, indemnities, procurement contracts, and domestic implementing laws determine whether and how those responsibilities are transferred to an operator. COPUOS debris-mitigation and long-term-sustainability guidelines provide important evidence of expected conduct but remain distinct from enforceable operator-specific obligations.

The FCC's 2022 Second Report and Order illustrates the importance of authorization-specific analysis. It created the first rule requiring covered non-geostationary operators to complete uncontrolled atmospheric-reentry disposal as soon as practicable and no later than five years after mission end, while addressing grandfathering and research-mission flexibility. The underlying licensing framework also

requires applicants to describe post-mission disposal plans and fuel reserved for disposal maneuvers. An accounting conclusion therefore depends on the effective date, license terms, object class, planned disposal method, and whether the operator can realistically avoid the required activity.

ESA's Zero Debris approach illustrates a different but complementary evidence layer. It emphasizes disposal-success probabilities above 90%, shorter LEO clearance, continuous monitoring of fuel and thrusters, passivation, and removal interfaces. NASA's active orbital-debris standard and procedural requirements similarly integrate debris planning into mission formulation. These standards are valuable inputs to ORDOM's engineering and cost scenarios. They create an accounting obligation only when incorporated into law, license, contract, funding condition, or a sufficiently specific constructive commitment under IFRS.

3.10 Relationship to MCDM, RLCAM, and NUIM

The four models answer different questions and cannot substitute for one another. MCDM determines reusable-launch consumption. RLCAM assigns that cost. NUIM tests recoverability at the appropriate economic unit. ORDOM identifies and measures retirement obligations. A favorable NUIM forecast cannot justify omission of a legal deorbit obligation, and a large ORDOM provision does not prove the network is impaired. The same engineering evidence can affect all four models, but each uses it for a distinct accounting objective.

While ORDOM belongs to an integrated space-accounting concept, it operates as a fully autonomous, modular framework. Entities do not need to implement MCDM, RLCAM, or NUIM to apply ORDOM. The model integrates seamlessly with standard straight-line depreciation under IAS 16 or ASC 360 and standard impairment testing under IAS 36 or ASC 360.

$$\text{MCDM} \rightarrow \text{RLCAM} \rightarrow \text{NUIM} \rightarrow \text{ORDOM} \tag{1}$$

4. Why Conventional Orbital-Retirement Accounting Can Fail

4.1 Engineering plans are mistaken for present obligations

A mission plan can reserve propellant and schedule a disposal maneuver without creating a liability if the entity retains a realistic alternative and no legal or constructive duty exists. The reverse can also occur: a license or contract can create an obligation even when the engineering team expects disposal to require no incremental cash outflow. Recognition begins with the obligation source, not the mission plan or budget.

4.2 Social externalities are recorded as if they were entity liabilities

Orbital debris imposes collision risk and congestion on other operators and future missions. Those externalities are economically important but are not automatically liabilities of the reporting entity. Recording the expected social cost of debris without an enforceable or constructive obligation would overstate liabilities and create inconsistent practice. ORDOM distinguishes stewardship metrics from financial-statement obligations.

4.3 Legal rights are ignored because exercise is considered remote

A counterparty or regulator may hold a right to require deorbiting even when management expects the right not to be exercised. Under US GAAP, a conditional legal asset retirement obligation can still require recognition at fair value if reasonably estimable. Historical Iridium disclosures demonstrate that a remote exercise probability can coexist with a recognized, probability-weighted ARO.

4.4 Asset derecognition is confused with liability settlement

Removing a satellite from service, writing its carrying amount to zero, or losing communications does not settle a disposal obligation. In fact, loss of control can increase expected removal cost. The asset and liability ledgers must therefore remain linked but distinct.

4.5 Natural decay is treated as costless without evidence

A satellite expected to reenter naturally can still require post-mission monitoring, collision-avoidance support, passivation, coordination, regulatory reporting, insurance, or design expenditure. Conversely, costs embedded in the original orbit selection and design may not create a future cash outflow. ORDOM separates sunk design cost, future operating cost, and present obligation.

4.6 High self-disposal probability excludes fallback scenarios

A 90% or 95% probability of successful self-disposal does not make the remaining probability irrelevant. If the entity remains obligated to arrange active removal, extended tracking, or another remedy after self-disposal failure, the fallback path belongs in the expected cash flows. If the operator has no obligation to fund that fallback, it should not be included merely because remediation is desirable.

4.7 Fleet aggregation shields object-specific duties

A constellation can be managed as a portfolio, but a license may attach disposal duties to each spacecraft. An expected-value portfolio estimate is appropriate only when obligations and cost drivers are sufficiently homogeneous and the model preserves object-level exceptions. A failed high-altitude satellite cannot be offset mechanically by many low-altitude objects that will decay naturally.

4.8 Discount rates become plugs

Long settlement horizons make provisions sensitive to discounting. Management can understate liabilities by adding non-performance risk or optimistic technology effects to the rate while also reducing cash-flow scenarios. ORDOM locks the risk treatment, separates real and nominal assumptions, and reconciles the rate to the applicable framework.

4.9 Voluntary zero-debris commitments are inconsistently treated

A public commitment can be ignored even after it becomes sufficiently specific to create a valid expectation under IFRS, or recognized prematurely when it remains aspirational. ORDOM requires evidence of scope, specificity, communication, established practice, affected counterparties, and the entity's realistic ability to withdraw.

4.10 Disclosure is fragmented

Operators may describe debris obligations in business or risk-factor sections, disclose satellite write-offs in property and equipment notes, and omit a coherent bridge to provisions, cash-flow timing, or critical estimates. Investors then cannot determine whether the absence of a liability reflects immateriality, lack of a present obligation, inability to estimate, or aggregation within another line item.

5. The Orbital Retirement, Decommissioning and Debris Obligation Model

5.1 Measurement objective and principles

ORDOM's objective is to identify and faithfully measure the present obligation associated with retiring an orbital object or related long-lived asset, using settlement scenarios that are complete, internally consistent, and supported by legal, contractual, engineering, operational, and market evidence. The model does not seek the broadest possible liability; it seeks the liability required by the applicable framework.

- **Obligation first.** Recognition is not driven by the existence or size of an expected disposal cost.
- **Past-event discipline.** Launch, operation, license acceptance, contractual activation, public commitment, or failure can be the obligating event only when the governing terms support that conclusion.
- **Unit integrity.** The unit of obligation follows the duty, not automatically the asset register, CGU, segment, or constellation.
- **Scenario completeness.** Self-disposal, partial failure, active removal, extended tracking, and alternative settlement methods are included when within the obligation.
- **Evidence hierarchy.** Legal instruments and observable technical data precede unsupported management assumptions.
- **No double-counting.** Risk is reflected once, through cash flows, probabilities, or rates as permitted by the framework.
- **Asset-liability separation.** Impairment and derecognition do not settle the obligation unless the entity is legally released.
- **Disclosure connectivity.** Recognized liabilities, unrecognized exposures, and operational metrics are presented in connected but distinct layers.

5.2 Core notation

Table 2. ORDOM notation.

Symbol	Definition
i	Individual spacecraft, launch-vehicle upper stage, hosted payload, gateway, or other long-lived asset.
g	Homogeneous obligation group or launch batch.
s	Settlement scenario, such as self-disposal, assisted disposal, active removal, natural decay, or graveyard-orbit transfer.
t	Reporting date.
$PE_{i,t}$	Past-event indicator.
$OA_{i,t}$	No-realistic-alternative or unavoidable-obligation indicator.
$OS_{i,t}$	Obligation-source indicator: legal, contractual, or constructive as permitted by the framework.
$R^{Ifrs}_{i,t}$ / $R^{Us}_{i,t}$	Recognition decision under IFRS or US GAAP.
$p_{i,s,t}$	Probability of settlement scenario s.
$C_{i,s,t}$	Nominal settlement cash flow in scenario s.
$\tau_{i,s,t}$	Expected time to settlement.
r^{Ifrs}_{t}	IFRS discount rate consistent with IAS 37.
$r^{carf}_{i,l}$	US GAAP credit-adjusted risk-free rate for liability layer l.
$P^{Ifrs}_{i,t}$	IFRS provision.
$ARO^{Us}_{i,t}$	US GAAP asset retirement obligation.
$ARC_{i,t}$	Associated capitalized asset retirement cost.
$T_{i,t}$	Telemetry and disposal-readiness vector.
$Q_{i,t}$	Probability of successful self-disposal.
$\Delta P_{i,t}$	Change in the recognized obligation.

5.3 Obligation taxonomy

Table 3. ORDOM obligation taxonomy.

Obligation family	Typical activity	Potential source	ORDOM treatment
Self-disposal	Orbit lowering, controlled reentry, graveyard-orbit transfer, passivation.	License, law, contract, or constructive commitment.	Recognize only after the present-obligation gate; include operations, propellant-related services, tracking, reporting, and execution costs.
Natural-decay compliance	Selecting disposal orbit and monitoring until reentry.	License conditions or mission authorization.	Include future monitoring and compliance costs if unavoidable; do not recreate sunk design cost as a liability.
Fallback servicing	Inspection, life extension, propulsion support, or assisted maneuver.	Contract, license remedy, or constructive commitment.	Include probability-weighted fallback when the entity bears the cost and cannot avoid settlement.
Active debris removal	Capture, tow, controlled reentry, or reorbit by a servicing provider.	Specific license, contract, indemnity, or valid constructive commitment.	Do not assume active-removal cost solely because an interface exists or service is technically possible.
Post-mission monitoring	Tracking, conjunction assessment, coordination, insurance, reporting.	License, regulation, contract.	Recognize future unavoidable costs when they are part of settling the obligation, not ordinary ongoing network operations.
Accident remediation	Debris characterization, cleanup, claims, penalties.	Accident-specific law, judgment, settlement, or contract.	Analyze separately under IAS 37/ASC 450/ASC 410-30 unless integral to the retirement obligation.
Launch-stage disposal	Upper-stage passivation, controlled reentry, disposal orbit.	Launch license, customer contract, government standard.	Separate from payload obligation unless one contract creates a combined settlement duty.

5.4 Present-obligation gate

For each object or homogeneous group, ORDOM records three threshold conditions: a qualifying past event, an obligation source recognized by the applicable framework, and the absence of a realistic alternative to settlement. The gate is deliberately binary at the recognition stage; probability and uncertainty enter the framework-specific recognition test and measurement model after the obligation is identified.

$$PO_{i,t} = PE_{i,t} \times OS_{i,t} \times OA_{i,t} \tag{2}$$

PO equals one only when all three conditions are supported. Under IFRS, OS can represent a legal or constructive obligation. Under US GAAP ASC 410-20, OS must represent a legal obligation. A published environmental aspiration can therefore satisfy neither framework, satisfy IFRS only, or satisfy both if incorporated into a binding license or contract.

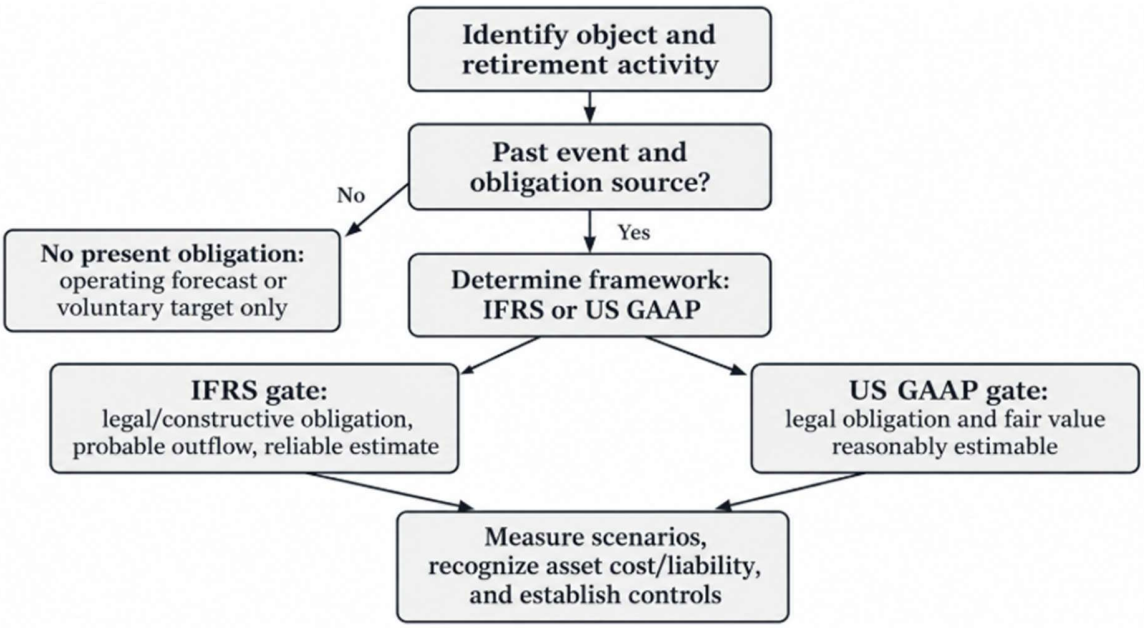


Figure 2. ORDOM recognition and measurement gate.

5.5 Framework-specific recognition functions

$$R_{i,t}^{\text{IFRS}} = \mathbf{1}\{PO_{i,t} = 1,\ \Pr(\text{Outflow}_{i,t}) > 50\%,\ \text{Estimate}_{i,t}\ \text{reliable}\} \tag{3}$$

$$R_{i,t}^{\text{US}} = \mathbf{1}\left\{\text{Legal obligation}_{i,t} = 1,\ \text{Fair value}_{i,t}\ \text{reasonably estimable}\right\} \tag{4}$$

The formulas are decision maps rather than replacements for the standards. Under IFRS, a possible obligation or an obligation with an outflow that is not probable is generally disclosed as a contingent liability unless remote. Under US GAAP, a conditional ARO can be recognized even when the exercise of a right is remote because fair value incorporates the probability of the settlement path. If fair value cannot be reasonably estimated, the entity discloses the obligation and the reasons for the estimation limitation and reassesses it each reporting period.

5.6 Unit of obligation

The unit of obligation is the smallest unit to which the enforceable or constructive duty attaches and for which settlement scenarios can be identified without arbitrary allocation. It can be an individual spacecraft, upper stage, hosted payload, launch batch, orbital plane, license population, or contractual fleet. The unit is not determined by the NUIM impairment unit. A global network can be one CGU while each spacecraft remains a separate unit of legal disposal responsibility.

Table 4. Determining the unit of obligation.

Fact pattern	Likely obligation unit	Reason
License requires disposal of each space station.	Individual spacecraft or homogeneous license cohort.	The duty is object-specific even if execution is managed as a fleet.
One launch contract requires the provider to dispose of the upper stage and hosted deployment module.	Launch-stage system or contract bundle.	Settlement activities are contractually integrated.
A mass-deorbit clause gives a counterparty the right to order retirement of the whole constellation.	Contractual constellation.	The trigger and settlement service attach to the fleet as a unit.
A public policy commits the operator to fund removal of any failed satellite in a named generation.	Generation cohort under IFRS, if a constructive obligation exists.	The commitment is defined by generation and failure condition.
Different altitudes, propellant systems, and license terms within one constellation.	Separate groups or individual objects.	Aggregation would hide materially different timing, method, and cost.

5.7 Settlement cost stack

ORDOM builds nominal scenario cash flows from a controlled cost stack. Cost categories are included only when necessary to settle the present obligation and consistent with the reporting framework's measurement objective. Internal costs use incremental or directly attributable resource consumption rather than arbitrary corporate overhead. Third-party costs use current market evidence, contracts, requests for information, or appropriately adjusted analogues.

$$C_{i,s,t} = C^{\text{ops}} + C^{\text{maneuver}} + C^{\text{passivation}} + C^{\text{tracking}} + C^{\text{reentry}} + C^{\text{servicing}} + C^{\text{removal}} + C^{\text{regulatory}} + C^{\text{other}} \quad (5)$$

Table 5. ORDOM settlement cost stack.

Cost component	Examples	Exclusion control
Mission operations	Flight dynamics, command windows, engineering, ground-station time.	Exclude ordinary network operations not required for settlement.
Maneuver execution	Orbit lowering, reorbiting, collision avoidance during disposal.	Do not assign sunk onboard propellant purchase as a future cash outflow; include future services and consumables actually paid.
Passivation	Battery discharge, tank venting, pressure relief, transmitter shutdown.	Exclude manufacturing cost already incurred unless a future service remains.
Tracking and coordination	Conjunction assessment, regulator notices, space-traffic coordination.	Limit horizon to the period required by the obligation.
Reentry safety	Controlled-reentry planning, maritime/aviation coordination, casualty-risk analysis.	Exclude unrelated launch or operational safety costs.
Servicing or removal	Inspection, docking, towing, capture, removal vehicle, launch service.	Include only when the entity bears the obligation and the active-removal amount passes the Section 5.9 estimability gate.
Regulatory and legal	License modification, filing, compliance review, settlement administration.	Exclude hypothetical penalties unless an obligating breach has occurred or the penalty is part of the settlement.
Insurance/reimbursement	Premiums required to settle or execute disposal.	Recognize reimbursement separately; do not net uncertain insurance proceeds against the liability.

5.8 Disposal scenario tree

A scenario tree begins with the planned disposal method and expands only for outcomes within the obligation's settlement perimeter. A self-disposal plan can include successful maneuver, partial propulsion failure requiring prolonged ground operations, loss of command requiring natural decay and monitoring, and active removal if the operator must fund it. A graveyard-orbit plan can include nominal reorbit, reduced propellant margin, failed passivation, and contracted servicing. The sum of scenario probabilities equals one for the defined obligation.

$$EC_{i;} \quad = \sum_{s=1}^{S} p_{i,s,t}\ C_{i,s,t}, \qquad \sum_{s=1}^{S} p_{i,s,t} = 1 \tag{6}$$

5.9 Self-disposal and fallback removal

Where self-disposal failure creates an unavoidable fallback duty, ORDOM expresses the expected cost as a nested path rather than a single success-rate adjustment. This makes the source of each probability transparent and prevents a high nominal success rate from excluding a low-probability, high-cost removal scenario.

$$EC_{i,t} = Q_{i,t}C_{i,t}^{\text{self}} + \left(1 - Q_{i,t}\right)\left[q_{i,t}C_{i,t}^{\text{assist}} + \left(1 - q_{i,t}\right)C_{i,t}^{\text{ADR}}\right] \tag{7}$$

C_{ADR} is included only when the entity has a present obligation to arrange or fund active removal and the cost passes the estimability gate below. The existence of a removal interface, a technically feasible servicing market, or a regulator's policy preference does not itself establish either responsibility or a measurable liability. If no present obligation exists, active-removal cost remains a risk-management scenario or sustainability metric rather than a financial-statement liability.

Estimability Gate for Active Debris Removal (C_{ADR}). To prevent arbitrary provision accounting, the active-removal component is subject to a strict measurement hierarchy linked to the ORDOM cost-source levels in Section 7.3:

1. ***Recognized provision / ARO:*** C_{ADR} is included in the recognized liability only when supported by ORDOM Level 1 or Level 2 evidence, such as binding service contracts, executed market quotes, or active procurement programs that make the settlement amount reliably estimable under IAS 37 or reasonably estimable at fair value under ASC 410-20.
2. ***Contingent liability / disclosure:*** If an active-removal duty exists but pricing cannot be estimated because of technological or commercial immaturity, C_{ADR} is excluded from the recognized provision or ARO. Under IAS 37, the unmeasurable component is disclosed as a contingent liability unless the possibility of outflow is remote; under ASC 410-20, the entity discloses the obligation and the specific reasons why fair value is not reasonably estimable.

The estimability gate affects measurement, not the existence of the underlying duty. The entity reassesses the excluded component at each reporting date and recognizes it prospectively once the applicable estimability threshold is satisfied.

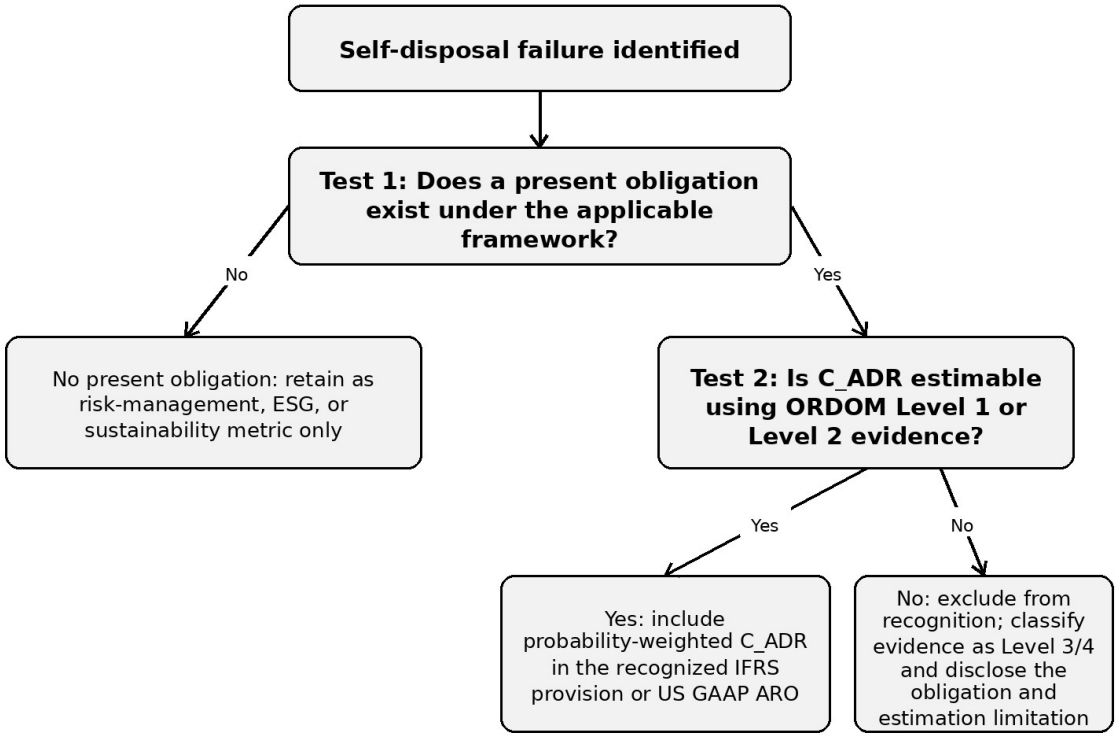


Figure 3. ORDOM estimability gate for active debris removal cost (C_ADR).

5.10 Telemetry and disposal-readiness vector

Telemetry informs measurement by updating the probability, timing, and method of settlement. ORDOM uses a disposal-readiness vector rather than a single health score because different components create different failure paths. Propellant margin affects maneuver capability; propulsion health affects thrust reliability; attitude control affects pointing; command-and-control affects execution; battery and power affect passivation; altitude and ballistic coefficient affect natural decay; and capture-interface readiness affects active-removal cost.

$$T_{i,t} = [\text{FUEL,PROP,ADCS,TTC,POWER,PASS,ALT,DRAG,CONJ,ADR}]_{i,t} \tag{8}$$

The vector is supported by controlled source data: telemetry archives, flight-dynamics models, fault logs, engineering change orders, disposal simulations, license submissions, insurer reviews, and independent mission-assurance analysis. Management cannot reduce a provision merely because an internal engineering team remains confident. The change must be tied to observable evidence and approved through the accounting-estimate governance process.

5.11 Probability update and evidence hierarchy

Rather than relying on an unobservable abstract function, ORDOM updates scenario probabilities using a Bayesian formulation based on engineering telemetry evidence:

$$p(S_s \mid T_{i,t}) = \frac{p(T_{i,t} \mid S_s) \cdot p_0(S_s)}{\sum_{k=1}^{S} p\left(T_{i,t} \mid S_k\right) \cdot p_0(S_k)} \tag{9}$$

Where $p_0(S_s)$ represents the baseline design probability for settlement scenario S_s, and $p(T_{i,t} \mid S_s)$ is the likelihood of observing telemetry vector $T_{i,t}$ given that settlement state. Telemetry therefore converts discrete engineering states—Nominal, Minor Anomaly, Loss of Redundancy, and Uncontrolled—into normalized, audit-verifiable posterior probability distributions rather than unsupported point estimates.

The posterior is not produced from telemetry in isolation. Current legal or license conditions (L), contractual terms (K), market evidence for settlement services (M), and historical performance for comparable missions (H) determine the admissible scenario set, inform priors and likelihood functions, and constrain cost and timing assumptions. First-order evidence consists of object-specific telemetry and binding instruments; second-order evidence includes demonstrated performance of the same satellite generation and executed servicing contracts; third-order evidence includes industry benchmarks and management estimates. Priors, likelihood functions, calibration cohorts, overrides, and posterior changes must be version-controlled and independently approved. Zero-probability assignments require compelling evidence because they can mechanically eliminate low-frequency fallback paths.

5.12 IFRS measurement engine

$$P_{i,t}^{\text{IFRS}} = \sum_{s=1}^{S} \frac{p_{i,s,t} C_{i,s,t}}{\left(1 + r_t^{\text{IFRS}}\right)^{\tau_{i,s,t}}} \tag{10}$$

Equation 10 is an expected-present-value representation. The selected technique must remain consistent with IAS 37's best-estimate objective. Cash flows are nominal or real consistently with the discount rate. Inflation is not duplicated. Non-performance risk is not inserted when already excluded by the standard or reflected elsewhere. Changes in settlement amount, timing, probabilities, and the current discount rate are remeasured at each reporting date and accounted for under IFRIC 1.

5.13 US GAAP measurement engine

$$ARO_{i,t}^{\text{US}} = \text{FV}\left(\left\{p_{i,s,t}, C_{i,s,t}, \tau_{i,s,t}\right\}, r_{i,l}^{\text{CARF}}, \text{Market risk}_{i,t}\right) \tag{11}$$

US GAAP's objective is fair value, not an entity-specific best estimate. Expected-cash-flow techniques incorporate possible settlement amounts and timing, market-participant assumptions, and the credit-adjusted risk-free rate for the relevant liability layer. When observable active-removal prices do not exist, ORDOM documents the valuation hierarchy, cost-build-up method, market-risk adjustment, technological maturity, service capacity, and sensitivity. A Level 3 engineering build-up does not by itself overcome ASC 410-20's reasonable-estimability threshold: if fair value remains not reasonably estimable, recognition of that component is deferred and the required FIN 47 disclosures are provided. A discounted amount is not labeled fair value merely because it has been present-valued.

5.14 Portfolio aggregation

Large constellations require scalable estimation. ORDOM permits portfolio measurement for homogeneous objects with common legal terms, design, altitude, disposal method, and cost drivers. The model retains object-level exception flags for failed propulsion, altered licenses, collision events, different orbital shells, or unusual contractual obligations. The portfolio amount is the sum of group estimates, not a top-down percentage of satellite book value.

$$P_t^{\text{portfolio}} = \sum_{g=1}^{G} N_{g,t} \sum_{s=1}^{S_g} \frac{p_{g,s,t} C_{g,s,t}}{(1 + r_t)^{\tau_{g,s,t}}} + \sum_{i \in \text{Exceptions}} P_{i,t} \tag{12}$$

5.15 Rollforward and reconciliation

$$P_t^{\text{close}} = P_t^{\text{open}} + P_t^{\text{new}} + P_t^{\text{revision}} + P_t^{\text{time}} - P_t^{\text{settled}} - P_t^{\text{released}} \tag{13}$$

Every movement is tagged to an object or obligation group and reconciled to the general ledger. New obligations arise from qualifying events. Revisions capture changes in amount, probability, timing, method, discount rates, or legal scope. Unwinding or accretion reflects passage of time. Settlement records actual expenditure. Release requires legal extinguishment, transfer, expiration, or another event that removes the present obligation; management's decision not to perform the activity is not a release.

5.16 Disposal-success shortfall

$$\text{Shortfall}_{i,t} = \max\left(0, Q_{i,t}^{\text{required}} - Q_{i,t}^{\text{supportable}}\right) \tag{14}$$

The shortfall is a governance indicator, not a liability amount. A decline below a regulatory or contractual success threshold can trigger a license response, higher fallback cost, an impairment indicator, a useful-life review, or a new obligation. Accounting recognizes only the resulting present obligation and measured settlement expenditure.

6. Recognition Architecture and Accounting Entries

Table 6. Comparative lifecycle journal entries under IFRS and US GAAP.

Lifecycle Stage	IFRS Entry (IAS 37 / IFRIC 1)	US GAAP Entry (ASC 410-20)
1. Initial Recognition *(At launch or other obligating event)*	Dr. Satellite Asset (ARC—Asset Retirement Cost) Cr. Provision for Orbital Retirement (IAS 37)	Dr. Satellite Asset (ARC) Cr. Asset Retirement Obligation (ARO)
2. Annual Time Effect *(Period-end)*	Dr. Finance Expense (Unwinding of Discount) Cr. Provision for Orbital Retirement	Dr. Accretion Expense Cr. Asset Retirement Obligation
3. Upward Estimate Revision *(Anomaly or regulatory change)*	Dr. Satellite Asset (ARC—subject to Impairment Ceiling) or P&L Cr. Provision for Orbital Retirement	Dr. Satellite Asset (ARC) or P&L if no future benefit remains Cr. ARO Layer (discounted at current credit-adjusted risk-free rate)
4. Settlement *(Deorbit execution)*	Dr. Provision for Orbital Retirement Cr. Cash / Servicing Provider Dr./Cr. Gain / Loss on Settlement	Dr. Asset Retirement Obligation Cr. Cash / Servicing Provider Dr./Cr. Gain / Loss on ARO Settlement

6.1 Identifying the obligating event

The obligating event is determined from the substance of the legal or constructive arrangement. Launch can be the event when placing the object in orbit immediately creates an unavoidable disposal duty. Orbital insertion can be the event when the license attaches only after successful deployment. Normal operation can create an obligation progressively when the duty depends on operating activity. A public commitment can create an IFRS constructive obligation only when the three cumulative criteria in Section 3.2—specificity, external reliance, and unavoidability—are all supported by evidence. Generic ESG ambitions, voluntary industry pledges, and board-approved intentions that remain realistically revocable do not pass the gate. A mission failure can create a new remediation obligation or change the settlement method of an existing obligation.

Table 7. Candidate obligating events.

Candidate event	Recognition analysis	Control evidence
Contract execution before construction	No ARO if the entity can cancel without unavoidable retirement activity; recognize when the duty is incurred.	Contract termination clauses, board authorization, cancellation rights.
Launch or orbital insertion	Common trigger when the license or contract makes disposal unavoidable once the object is placed in orbit.	License condition, mission authorization, launch manifest, insertion confirmation.
Operation over time	Use when the obligation is created by the activity performed, not merely by ownership.	Statutory formula, usage data, operating period.
Specific public commitment	Potential IFRS constructive obligation only when specificity, external reliance, and unavoidability are cumulatively satisfied; otherwise no provision arises from the statement alone.	Published asset or cohort scope, formal stakeholder communication, performance history, withdrawal consequences, and legal review.
Mission anomaly	May revise an existing obligation or create a new accident-remediation duty.	Telemetry, anomaly board report, regulator notice, legal opinion.
End-of-service decision	Can trigger impairment/derecognition; creates a liability only if it activates a present retirement duty.	Board decision, customer termination, license provisions, disposal plan.

6.2 Initial recognition under IFRS

When the obligation is incurred and the IAS 37 recognition criteria are met, the entity recognizes a provision and, when appropriate, includes the same amount in the cost of the related satellite, upper stage, or other asset. If the obligation arises while an asset is under construction, the retirement cost enters construction in progress and is depreciated when the asset becomes available for use. If the obligation relates to inventories produced during a period, IAS 2 interaction requires separate analysis.

Illustrative IFRS entry at launch:

Dr. Satellite asset—capitalized orbital retirement cost;
Cr. Provision for orbital retirement and debris obligations.

The amount is the IAS 37 best estimate at the obligating date, not the nominal maximum cost or a percentage of construction cost.

6.3 Initial recognition under US GAAP

When a legal ARO is incurred and fair value can be reasonably estimated, the entity recognizes the liability and increases the carrying amount of the related long-lived asset. The asset retirement cost is allocated systematically over the asset's useful life. If an existing asset is acquired with an ARO, the acquisition accounting and fair-value measurement must preserve the gross liability and related asset basis rather than netting expected disposal proceeds or insurance.

Illustrative US GAAP entry at launch:

Dr. Satellite asset—asset retirement cost;
Cr. Asset retirement obligation.

Subsequent periods record depreciation of the ARC and accretion of the ARO as separate expense effects.

6.4 Later-arising obligations

A new five-year disposal rule, license modification, or sufficiently specific zero-debris commitment can create an obligation after the asset has entered service. Under IFRS, IFRIC 1 generally adjusts the related asset under the IAS 16 cost model, but ORDOM applies the Impairment Ceiling Rule: capitalization is limited to the amount supportable under IAS 36, and any increase that cannot be absorbed by a recoverable asset is recognized immediately in profit or loss. If the spacecraft has already been fully impaired, derecognized, abandoned, or has reached the end of its useful life, the entire debit is an operational decommissioning expense.

Under US GAAP, an upward revision creates a new ARO layer discounted at the current credit-adjusted risk-free rate. The corresponding ARC is capitalized and depreciated over the remaining useful life only when future economic benefit remains. If no recoverable asset base remains, the debit is recognized in current earnings; otherwise the resulting asset or asset group remains subject to ASC 360 recoverability and impairment measurement.

6.5 Derecognition and settlement

Retirement from commercial service, physical reentry, transfer to a compliant disposal orbit, legal release, and completion of required passivation are separate milestones. The liability is settled only to the extent the obligation has been discharged. A satellite transferred to a graveyard orbit can be fully retired for accounting purposes even though the physical object remains in space, provided the enforceable disposal requirements have been satisfied. Conversely, a satellite written off after loss of communications can retain a substantial removal or monitoring liability.

6.6 Insurance and indemnities

Insurance can cover third-party liability, mass-deorbit services, launch failure, in-orbit anomalies, or removal cost. ORDOM distinguishes coverage for the asset from coverage for settlement of the obligation. Expected proceeds are recognized separately only when the applicable recognition threshold is met. A policy exclusion, deductible, insurer credit risk, or annual renewal requirement can materially affect the reimbursement asset without changing the gross obligation.

7. Measurement Architecture and Anti-Management-Override Controls

7.1 Legal-scope matrix

Before cost modeling, legal and regulatory teams complete a scope matrix for each object or cohort. The matrix identifies the launching state, licensing authority, operator, owner, contractual indemnitors, disposal method, deadline, success threshold, reporting duties, insurance requirements, and consequences of nonperformance. Accounting signs off only after the source of the obligation and the obligating event are documented.

7.2 Engineering-to-accounting bridge

Engineering models often report probability of successful disposal, post-mission lifetime, casualty risk, propellant margin, and conjunction risk. Accounting models require expected cash outflows, timing, and obligation scope. ORDOM bridges the two through a controlled mapping. A 90% disposal-success estimate is not multiplied directly by asset cost. Instead, each failure mode is assigned to a settlement path, the entity's responsibility for that path is evaluated, and only then are the associated cash flows measured.

7.3 Cost estimation hierarchy

- **Level 1—executed prices.** Existing servicing contracts, supplier quotes, regulator fees, ground-station rates, and actual disposal missions.
- **Level 2—observable analogues.** Comparable spacecraft classes, launch services, insurance premiums, and demonstrated mission-operations costs adjusted for scale and timing.
- **Level 3—engineering build-up.** Labor hours, command windows, mission design, servicing vehicle mass, launch allocation, and risk contingencies supported by independent review.
- **Level 4—management estimate.** Used only when other evidence is unavailable, with explicit uncertainty ranges and a plan to replace the estimate as evidence develops.

For C_{ADR}, only Level 1 or Level 2 evidence passes the recognition-measurement gate in Section 5.9. Level 3 or Level 4 estimates may support internal risk analysis, sensitivity, and disclosure, but do not enter the recognized provision or ARO until the amount becomes reliably estimable under IAS 37 or reasonably estimable at fair value under ASC 410-20.

7.4 Technology-cost assumptions

Active debris removal and in-orbit servicing markets are developing. Forecast cost declines cannot be included merely because technology is expected to mature. Under IAS 37, future events are reflected when sufficient objective evidence exists. Under ASC 410-20 fair value, market participants may consider expected technological change, but the valuation must reflect risk, capacity, service availability, and the obligation's settlement date. ORDOM uses observable procurement programs, executed demonstration missions, credible supplier capacity, and launch-market evidence rather than a generic learning curve.

7.5 Inflation and currency

Settlement may be denominated in US dollars, euros, or another currency and can involve launch services whose prices respond differently from general inflation. Nominal cash flows use component-specific inflation and a nominal discount rate. Real cash flows use real rates. Foreign-currency provisions are retranslated under the applicable foreign-currency guidance, while the underlying asset-retirement-cost adjustment follows IFRIC 1 or ASC 410-20 as appropriate. Currency volatility is not hidden in a blended contingency percentage.

7.6 Discount-rate governance

The controller documents the rate source, term, currency, inflation basis, and risk treatment. IFRS rates are updated at each reporting date under current IAS 37 and IFRIC 1. US GAAP liability layers preserve the applicable credit-adjusted risk-free rates for accretion and downward revisions, with current rates for upward layers. The model prevents management from resetting all historical US GAAP layers to current rates or retaining stale IFRS rates.

7.7 Double-counting controls

Table 8. ORDOM double-counting controls.

Risk	Permitted location	Prohibited duplication
Self-disposal failure	Scenario probability or cost contingency.	Additional discount-rate premium for the same failure.
Launch price uncertainty for removal mission	Probability-weighted supplier cost or market-risk adjustment under fair value.	Generic contingency plus inflated launch quote plus rate premium.
Non-performance risk	Only as permitted by the applicable fair-value or provision framework.	Reduction in both cash flows and discount rate.
Inflation	Nominal cash flows with nominal rate, or real cash flows with real rate.	Inflation in both escalation and discount spread.
Legal uncertainty	Recognition assessment and, when appropriate, scenario probabilities.	Artificially reducing both recognition probability and settlement cost after recognition.

7.8 Management-override indicators

- A decrease in the provision immediately before a financing, covenant test, or compensation measurement without new legal, market, or telemetry evidence.
- A change from individual-object measurement to fleet averaging after a high-cost failure.
- A disposal-success probability that increases while propulsion health, command capability, or propellant margin deteriorates.
- A switch from active-removal fallback to natural decay without a license modification or legal release.
- An impairment write-off accompanied by release of the liability even though the duty remains.
- Use of a higher discount rate without a corresponding change in the applicable framework or liability characteristics.

- Exclusion of internal operations costs despite a requirement to maintain ground support until disposal completion.
- Inclusion of reputational or societal costs to create an earnings reserve not tied to a present obligation.

7.9 Back-testing

Each completed disposal is back-tested against predicted method, timing, operations hours, supplier cost, and settlement amount. Model bias is evaluated by satellite generation and orbit. Persistent underestimation requires recalibration and can indicate a control deficiency. Persistent overestimation can signal prudence bias or deliberate earnings smoothing. Back-testing results are reported to the audit committee or equivalent governance body.

8. Public Evidence and Practical Applications

8.1 Evidence protocol

Public filings rarely provide enough information to compute a company-specific ORDOM liability. The examples below therefore follow a strict protocol: observed facts are stated as disclosed; accounting implications are framed as questions or model applications; numerical illustrations use hypothetical assumptions; and no conclusion is drawn about whether the named entity should recognize a different amount.

8.2 Iridium: current license obligations and historical ARO precedent

Iridium's 2025 Form 10-K describes an architecture of 66 operational LEO satellites in six orbital planes, together with in-orbit spares and ground infrastructure. It states that all second-generation satellites are subject to a 25-year de-orbit standard under the FCC authorization of the current constellation. This is important because it demonstrates that the FCC's later five-year rule does not eliminate authorization-specific and grandfathered disposal terms. ORDOM therefore begins with each operator's actual license rather than a universal benchmark.

Historical Iridium filings provide a rare public accounting precedent. The 2014 Form 10-K stated that the US government, Boeing, and Motorola Solutions held specified rights to require mass deorbit of the first-generation constellation. If a mass deorbit were required, Iridium disclosed a potential payment to Boeing of US$18.0 million plus an amount equivalent to a US$2.5 million insurance premium. Iridium concluded that the rights met the definition of an asset retirement obligation, assessed exercise as remote, and recorded an ARO of approximately US$0.2 million. The case shows why a conditional legal right can produce a recognized probability-weighted liability even when management considers exercise remote (Iridium Communications, 2015).

ORDOM application—Iridium

The current 25-year license obligation and the historical contractual mass-deorbit ARO should be analyzed as distinct obligation populations. The public record demonstrates legal scope and accounting precedent, but it does not disclose the current asset-level cost model, probabilities, or whether any historical obligation remains. ORDOM would preserve those distinctions rather than infer one liability from the other.

8.3 Spire Global: decommissioning loss versus retirement liability

Spire Global reported losses of US$9.1 million and US$3.4 million on decommissioned satellites and other asset write-offs for 2025 and 2024. The 2025 amount was primarily attributed to the decision to stop supporting three underperforming satellites, deorbiting three additional satellites, and disposal of obsolete construction-in-process materials. The 2024 amount was primarily associated with failure to establish communications with multiple satellites (Spire Global, 2026).

The disclosure is a useful boundary example. A write-off measures loss of future economic benefit in the asset. ORDOM asks separately whether stopping support, completing deorbit, or losing communications changes an existing disposal obligation. A completed compliant deorbit can settle the liability. A loss of communications can increase it if active removal or prolonged monitoring becomes unavoidable. Public

information does not reveal Spire's license terms or obligation measurement, so the paper draws no conclusion about its liability accounting.

ORDOM application—Spire

Asset decommissioning expense is not a proxy for the orbital retirement provision. The two amounts can move in the same direction, opposite directions, or at different dates. A control reconciliation should explain the effect of each decommissioned object on both the asset register and obligation register.

8.4 Sirius XM: a completed deorbit milestone

Sirius XM's 2025 Form 10-K states that XM-3 was removed from service during the first quarter of 2025 and that the deorbiting process was completed in November 2025. This fact pattern illustrates the distinction between retirement decision, execution period, and settlement date. ORDOM would identify when the relevant obligation was incurred, accrete or unwind the liability through the execution period, recognize actual deorbit costs against the provision, and record a settlement difference when the required activity was completed (Sirius XM Holdings, 2026).

The public filing does not disclose an ARO amount or the deorbit cost. Accordingly, the example is operational rather than a conclusion about recognition. Its value is pedagogical: disposal is a process with multiple accounting dates, not a single event coincident with the last depreciation charge.

8.5 FCC five-year disposal: authorization-specific LEO application

The FCC's 2022 order requires covered non-geostationary stations ending their mission in, or passing through, LEO and planning uncontrolled atmospheric reentry to complete disposal as soon as practicable and no later than five years after mission end. The licensing framework requires a description of post-mission disposal plans and the amount of fuel reserved for disposal maneuvers. The rule also contains grandfathering and flexibility provisions. ORDOM therefore treats the five-year horizon as a legal input only for objects within scope and only after the relevant authorization becomes effective (FCC, 2022).

The FCC's 2022 rule can also create a direct present-value effect when it legally accelerates the settlement horizon of an already recognized obligation. Holding nominal disposal cost (C) constant and using a 5% discount rate, shortening the horizon from 25 years to 5 years removes 20 years of discounting:

$$PV_{25} = \frac{C}{(1.05)^{25}} \approx 0.295C, \qquad PV_5 = \frac{C}{(1.05)^5} \approx 0.784C, \qquad \frac{PV_5}{PV_{25}} - 1 \approx 165.8\%$$

Accordingly, the recognized present value increases by approximately 165.8% even though the nominal disposal cost is unchanged. The result applies only when the enforceable settlement date for an in-scope authorization is actually accelerated; grandfathering, license-specific transition provisions, changes in nominal cost, and probability revisions must be modeled separately. All else equal, the resulting liability increase raises balance-sheet leverage and can materially affect debt-to-equity ratios for LEO operators without any change in physical satellite design.

For an in-scope launch batch, the five-year deadline affects expected settlement timing, post-mission monitoring, collision-avoidance exposure, and the feasibility of natural decay. If orbital insertion altitude makes five-year decay supportable without a terminal maneuver, future cash outflows may be limited to monitoring and compliance. If a maneuver is required, the cost stack includes ground operations and execution services. If failed maneuver capability leaves the operator responsible for removal, the fallback scenario can be material.

8.6 ESA Zero Debris: success thresholds and active-removal interfaces

ESA's Zero Debris approach states that self-disposal should be verified before launch with a probability of success above 90%, that critical functions such as fuel and thrusters should be monitored continuously, and that missions should include interfaces enabling removal if self-disposal fails. ESA's

newer requirements also reduce the LEO disposal phase to a maximum of five years and apply more stringent requirements to constellations (ESA, 2023, 2025).

ORDOM uses these requirements as engineering and contractual evidence when applicable. A 90% threshold does not mean that 10% of replacement cost is the liability. The remaining probability must be mapped to specific settlement paths and legal responsibility. An active-removal interface can reduce expected capture cost, but it does not establish that the operator must pay for removal. The accounting conclusion follows the mission's contract, funding arrangement, authorization, and commitments.

8.7 Illustrative LEO portfolio measurement

Assume, solely for illustration, that a company launches a homogeneous batch of 100 LEO satellites. Launch creates a present obligation to complete disposal. Management supports an eight-year expected settlement horizon under the applicable authorization. The scenario model estimates a 91% probability of nominal self-disposal costing US$0.35 million per satellite, a 7% probability of assisted disposal costing US$1.20 million, and a 2% probability of contracted active removal costing US$7.50 million. Assume a 4.0% discount rate for this simplified IFRS illustration.

Table 9. Illustrative LEO scenario model—not company data.

Scenario	Probability	Nominal cost per satellite	Probability-weighted cost
Nominal self-disposal	91%	US$0.35m	US$0.3185m
Assisted disposal	7%	US$1.20m	US$0.0840m
Active removal	2%	US$7.50m	US$0.1500m
Total	100%		US$0.5525m

The expected undiscounted amount is US$55.25 million for the batch. Discounted for eight years at 4.0%, the simplified present value is approximately US$40.37 million. The model does not assert that these costs or probabilities are representative of any operator. It demonstrates why a low-probability removal scenario can comprise a material share of the liability and why the amount cannot be derived from satellite construction cost.

8.8 Illustrative telemetry-driven revision

Assume that a common propulsion anomaly affects ten satellites from the batch. Updated evidence reduces the self-disposal probability to 70%, increases assisted disposal to 20%, and increases active removal to 10%. Revised nominal costs are US$0.45 million, US$1.50 million, and US$8.00 million respectively; the expected settlement horizon becomes five years and the applicable simplified IFRS rate becomes 3.5%. The revised present value for the ten objects is approximately US$11.91 million, compared with approximately US$4.04 million under the original model, an increase of approximately US$7.88 million.

Under IFRS, the increase is accounted for under IFRIC 1, with the asset adjustment limited by the related asset's remaining carrying amount and subject to impairment review. If the affected satellites have already been written off, the increase is recognized in profit or loss. Under US GAAP, the upward revision forms a new liability layer discounted at the current credit-adjusted risk-free rate, with the related asset retirement cost depreciated over the remaining useful life if future benefit remains.

8.9 Illustrative GEO graveyard-orbit obligation

Consider a single GEO satellite with a present legal obligation to reorbit and passivate at the end of a 12-year remaining service period. Hypothetical scenarios are: 94% nominal reorbit and passivation at US$3.0 million; 5% partial failure requiring extended operations or servicing at US$12.0 million; and 1% failed disposal requiring contracted removal at US$40.0 million. The expected undiscounted amount is US$3.82 million. At a simplified 4.5% rate for 12 years, present value is approximately US$2.25 million.

This example shows that graveyard-orbit obligations differ from LEO reentry. The relevant costs can include reserved operations capability, reorbit maneuver design, passivation, post-maneuver verification, and regulatory reporting. A failed disposal path is included only if the operator remains responsible for remediation. The expected satellite residual value is not netted against the obligation.

9. Disclosure Architecture

9.1 Why a dedicated disclosure is preferable

Orbital obligations should not be dispersed solely across sustainability reports, risk factors, property-and-equipment notes, and generic provision tables. When material, a dedicated financial-statement note titled 'Orbital retirement and debris obligations' provides the clearest connection among the obligation source, related assets, carrying amount, settlement horizon, measurement uncertainty, and cash-flow consequences. The note can cross-reference detailed operational sustainability reporting without using that reporting as a substitute for recognition and measurement disclosures.

9.2 Three-layer disclosure model

Table 10. ORDOM three-layer disclosure architecture.

Layer	Objective	Core content
1. Financial-statement note	Explain recognized liabilities and their movement.	Accounting policy; obligation description; carrying amount; rollforward; timing; discount rates; reimbursements; settlement gains/losses; current/noncurrent classification.
2. Critical judgments and estimates	Explain uncertainty and management decisions.	Obligating event; constructive-obligation assessment; unit of obligation; scenario probabilities; cost sources; technology assumptions; sensitivity; estimation limitations.
3. Operational and sustainability reporting	Explain physical performance and stewardship.	Objects subject to disposal plans; disposal method; success-rate bands; completed disposals; failures; active-removal readiness; collision-avoidance and passivation metrics.

9.3 Minimum financial-statement note

- A description of the nature and legal or constructive source of material obligations, including affected orbit and asset class.
- The accounting policy under IAS 37 and IFRIC 1 or ASC 410-20, including the treatment of associated asset retirement cost and accretion or discount unwinding.
- A rollforward separating new obligations, revisions in cost, revisions in timing or probability, discount-rate effects, accretion or unwinding, settlements, and releases.
- Expected settlement timing, preferably in maturity bands rather than a single weighted-average date when outcomes are dispersed.
- Key measurement assumptions and the source hierarchy for cost, disposal success, and active-removal services.
- Material reimbursement rights, insurance, indemnities, and decommissioning funds presented separately.
- The relationship between decommissioned or impaired assets and changes in the obligation.
- Contingent liabilities or obligations not recognized because an outflow is not probable or fair value cannot be reasonably estimated, as applicable.
- Sensitivity to a reasonably possible change in cost, timing, self-disposal probability, and discount rate when material.

9.4 Proposed rollforward

Table 11. Illustrative ORDOM rollforward.

Movement	Recognized provision/ARO	Related asset retirement cost	Profit or loss effect
Opening balance	Opening liability	Opening ARC balance	—
New obligations incurred	Increase	Increase	Future depreciation
Revisions—amount/probability	Increase/(decrease)	Adjustment subject to framework	Immediate effect when asset adjustment is unavailable
Discount-rate change—IFRS	Increase/(decrease)	Adjustment under IFRIC 1	Potential immediate effect
Accretion/unwinding	Increase	—	Finance/accretion expense
Settlements	Decrease	—	Gain/loss if actual differs
Legal releases/transfers	Decrease	Decrease or profit/loss	As applicable
Closing balance	Closing liability	Closing ARC balance	—

9.5 Operational metrics that improve understanding

Operational metrics are useful when they explain the liability without revealing mission-sensitive information. Appropriate aggregated metrics can include the number of active objects subject to a disposal obligation, the number in disposal phase, disposal method by broad category, weighted-average settlement horizon, success-probability bands, completed disposals, and objects with unresolved anomalies. Exact propellant reserves, command frequencies, conjunction windows, and servicing-interface details can be omitted or aggregated for security and commercial reasons.

Layer 3 also provides a conceptual bridge to investor-focused sustainability reporting under IFRS S1 General Requirements for Disclosure of Sustainability-related Financial Information and IFRS S2 Climate-related Disclosures, issued by the International Sustainability Standards Board, and to the European Union's Corporate Sustainability Reporting Directive (CSRD). ORDOM does not convert sustainability objectives into accounting liabilities; rather, it supplies the governance architecture needed to connect reported orbital-stewardship metrics to controlled obligation registers, authorization-specific legal analysis, version-controlled telemetry, scenario assumptions, cost-source evidence, management approvals, and audit trails. This traceability allows preparers and assurance providers to distinguish enforceable retirement duties, unrecognized but decision-useful exposures, and voluntary zero-debris commitments, thereby reducing the risk of 'space-washing'—the presentation of broad orbital-sustainability claims without verifiable operational evidence, accounting boundaries, or reconciliation to recognized amounts and disclosed contingencies.

9.6 Sensitivity disclosure

$$\text{Sensitivity}_x = P(\theta_x + \Delta\theta_x) - P(\theta_x) \tag{15}$$

Sensitivity should focus on assumptions that management can support and users can understand. For example, the entity can disclose the effect of a five-percentage-point reduction in self-disposal probability, a one-year delay in settlement, a 10% increase in active-removal cost, or a 50-basis-point movement in the IFRS discount rate. A sensitivity is not a substitute for recording the best estimate; it explains the range of reasonably possible outcomes.

9.7 Disclosure of unrecognized obligations

The absence of a recognized amount can be decision-useful only when the reason is clear. Under IFRS, the entity distinguishes no present obligation, non-probable outflow, remote exposure, and inability to measure reliably. Under US GAAP, it distinguishes no legal obligation, fair value not reasonably estimable, and an obligation outside ASC 410-20. If estimation is impracticable because no active-

removal market exists, the entity describes the facts, possible settlement methods, and milestones expected to improve estimability.

9.8 Connected disclosure with NUIM

ORDOM and NUIM disclosures should reconcile but remain separate. The obligation note explains the liability and related asset-retirement-cost component. The impairment note explains the CGU or asset group, indicators, recoverable amount or fair value, and loss. A common asset cohort identifier allows users to understand why a propulsion anomaly increased the retirement provision, shortened useful life, and triggered impairment without treating those effects as one blended estimate.

10. Internal Controls, Audit Evidence, and Governance

10.1 Governance model

ORDOM requires joint ownership by accounting, legal, licensing, engineering, mission operations, procurement, insurance, and internal audit. No single function controls the full estimate. Legal determines obligation source and release. Engineering determines disposal capability and failure modes. Procurement and finance estimate cost. Accounting applies recognition and measurement. The audit committee oversees significant judgments and changes.

10.2 Minimum control set

Table 12. ORDOM minimum internal controls.

Control	Frequency	Evidence
Object-to-obligation register reconciliation	Quarterly and at each launch/disposal	Asset register, launch manifest, licenses, contracts, orbital catalog.
Legal obligation review	At launch, license change, contract change, anomaly, and reporting date	Legal memorandum, regulator correspondence, contract matrix.
Telemetry and disposal-readiness review	Continuous operational monitoring; formal quarterly close snapshot	Telemetry archive, flight-dynamics report, anomaly logs, propulsion and power health.
Cost-source refresh	At least annually and upon market/contract change	Supplier quotes, executed contracts, labor rates, launch-service evidence.
Probability calibration and back-test	Annually and after each disposal event	Baseline priors, likelihood functions, posterior distributions, historical outcomes, calibration cohorts, override log, and independent engineering review.
Discount-rate approval	Each reporting date under IFRS; new/revised layers under US GAAP	Treasury data, valuation memo, currency and term reconciliation.
Rollforward and general-ledger reconciliation	Each close	Subledger, journal entries, settlement invoices, asset records.
Disclosure committee review	Quarterly/annual reporting	Draft note, risk factors, sustainability metrics, consistency checklist.

10.3 Audit evidence

Audit evidence should include the underlying license and contract terms, external legal analysis when material or complex, technical disposal plans, telemetry extracts, management's probability methodology, third-party cost evidence, discount-rate support, and the reconciliation to the asset register. Representations are not substitutes for evidence. When the obligation depends on emerging active-removal technology, the auditor evaluates specialist competence, market evidence, model risk, and management bias.

10.4 Use of specialists

Aerospace engineers can evaluate post-mission lifetime, propulsion reliability, capture feasibility, casualty risk, and disposal-success probability. Valuation specialists can evaluate expected present value and market-participant assumptions. Legal specialists can interpret licenses, indemnities, launching-state arrangements, and constructive commitments. The controller remains responsible for integrating specialist work into the accounting framework and ensuring that assumptions are consistent across depreciation, impairment, insurance, and provision models.

10.5 Cybersecurity and data integrity

Telemetry is mission-critical and may be security sensitive. ORDOM should consume controlled summary outputs rather than unrestricted raw feeds. Data lineage documents the source system, extraction date, transformation, missing data, overrides, and approval. Changes to disposal probabilities require traceable links to engineering evidence. Access to model parameters is segregated from approval of journal entries.

10.6 Model validation

Independent validation evaluates mathematical accuracy, completeness of settlement paths, probability calibration, cost-source reliability, rate consistency, and sensitivity. Validation also tests whether a simpler model would provide equally faithful representation. Complexity is not a virtue. An object with a fixed contractual disposal price and date does not require a multi-factor telemetry model; a diverse constellation with uncertain fallback removal may.

11. Comparative IFRS and US GAAP Implementation

Table 13. IFRS-US GAAP comparison.

Topic	IFRS	US GAAP	ORDOM implementation
Scope of obligation	Legal and constructive obligations under IAS 37.	Legal obligations only under ASC 410-20; a constructive commitment without legal basis is outside ARO scope.	Maintain separate legal/contractual/constructive source flags; never import an IFRS-only constructive obligation into the US GAAP ledger.
Recognition threshold	Present obligation from a past event; probable outflow; reliable estimate.	Legal ARO recognized when incurred if fair value is reasonably estimable; conditional timing or method does not eliminate the duty.	Run two recognition gates on one controlled legal, contractual, and engineering fact base.
Initial measurement objective	Best estimate of settlement expenditure under IAS 37; expected value or most likely outcome, as appropriate.	Fair value under ASC 410-20, generally using expected present value and market-participant assumptions.	Use separate measurement engines; do not blend IAS 37 best estimate with ASC 410-20 fair value.
Discount rate	A current pre-tax rate reflecting time value and liability-specific risks not already reflected in cash flows; remeasured at each reporting date.	A credit-adjusted risk-free rate (CARF) assigned by liability layer; historical layers are preserved, while upward revisions use a current CARF.	Maintain dual rate subledgers and explicit controls against risk double-counting.
Associated asset (ARC)	Initial obligation generally capitalized under IAS 16 when criteria are met.	Asset retirement cost capitalized as part of the tangible long-lived asset.	Link each liability or homogeneous obligation group to the relevant object/component and depreciation schedule.

Subsequent revisions	IFRIC 1 generally adjusts the related asset under the cost model; excess reductions and post-useful-life changes enter profit or loss.	ASC 410-20 layering: upward revisions create new layers at a current CARF; downward revisions use the rate associated with the related prior layer.	Tag each revision to legal scope, cost, probability, timing, method, telemetry, or rate changes.
Passage of time	Unwinding of the discount recognized as a finance cost under the applicable presentation policy.	Accretion expense recognized using the rate assigned to each ARO layer.	Present unwinding/accretion separately from operating disposal and servicing costs.
Impairment and derecognition	Asset impairment or derecognition does not extinguish the provision. Under IFRIC 1 and IAS 36, an upward liability revision is capitalized only to the extent recoverable; otherwise it is recognized immediately in profit or loss.	Asset impairment or derecognition does not extinguish the ARO. The related ARC is subject to ASC 360; if no future economic benefit remains, the debit is expensed.	Apply the ORDOM Impairment Ceiling Rule and reconcile the asset register separately from the obligation register.
Settlement and release	Derecognize the liability only when settled, legally released, transferred, expired, or otherwise extinguished.	Derecognize the ARO upon settlement or legal extinguishment; recognize any settlement gain or loss.	Do not treat retirement from service, loss of communication, or an asset write-off as liability release.
Reimbursements and indemnities	Recognize a separate reimbursement asset when receipt is virtually certain, capped at the provision.	Apply the relevant separate-recognition guidance; gross ARO remains unless the entity is legally released.	Maintain independent obligation and reimbursement registers; prohibit unsupported netting.
Disclosure	IAS 37 class disclosures, movements, timing, uncertainties, major assumptions, and reimbursements; contingent liabilities when applicable.	ASC 410 descriptions and rollforward; disclose inability to estimate fair value and the reasons when applicable.	Use the ORDOM three-layer architecture: financial-statement note, critical judgments/estimates, and controlled operational sustainability metrics.

11.1 Dual-reporting ledger architecture

A dual reporter should maintain one operational obligation register and two accounting measurement layers. The operational register stores object identity, license, contract, disposal method, telemetry, scenario probabilities, nominal cost, and settlement timing. The IFRS layer applies constructive-obligation scope, probability recognition, current discounting, and IFRIC 1. The US GAAP layer applies legal-obligation scope, fair value, credit-adjusted risk-free layers, and ASC 410-20. Reconciliation differences are explainable by rule rather than by parallel spreadsheets with inconsistent source data.

$$\text{Operational facts} \rightarrow \text{IFRS engine} \mid \text{US GAAP engine} \tag{16}$$

11.2 Transition and new regulation

When a regulation changes, the entity first determines which objects and authorizations are in scope and the effective obligating date. Grandfathered objects remain under prior terms unless modified. New

obligations are recognized when incurred, not automatically at the announcement date. Existing obligations are remeasured for changes in timing, method, or scope. Disclosure explains the effect of regulation separately from ordinary telemetry and cost revisions.

11.3 Tax and rate-regulated effects

Tax deductibility, deferred tax, regulated recovery, and government reimbursement are outside ORDOM's core recognition model but can materially affect financial reporting. A tax deduction at settlement does not reduce the pre-tax provision; it affects IAS 12 or ASC 740. A regulator's promise to reimburse disposal cost can create a separate regulatory or reimbursement asset only when the relevant recognition criteria are met.

Under IAS 12 (as amended for single-transaction deferred tax), recognizing an initial asset retirement obligation (ARO) and corresponding asset retirement cost (ARC) gives rise to equal and offsetting deferred tax assets (DTA) and deferred tax liabilities (DTL) at inception, as the initial recognition exemption no longer applies. These positions unwind asynchronously over the asset's useful life and settlement horizon.

12. Normative Architecture and Future Empirical Agenda

12.1 Normative status

ORDOM is framed as a Design-Science Research (DSR) contribution in financial accounting. Rather than attempting a descriptive study of current reporting practice—which remains fragmented and inconsistent across aerospace filers—ORDOM provides a normative blueprint. Its purpose is to equip standard setters (IASB and FASB), regulators (SEC and FCC), auditing firms, and preparers with operational tools for objective, verifiably audited orbital provisions.

ORDOM is a normative accounting architecture. It does not claim empirical validation with proprietary object-level legal instruments, telemetry, settlement costs, or recognized ARO records. Public evidence establishes that disposal rules differ by authorization, that self-disposal success is a measurable engineering objective, that active removal is becoming a defined fallback strategy, and that operators recognize decommissioning losses and, in at least one historical public case, a satellite-related ARO. It does not establish the correct liability for any named entity.

12.2 Testable propositions

- **P1—Recognition timeliness.** Operators with explicit object-level license or contract obligations will recognize retirement liabilities earlier than operators relying only on nonbinding guidelines, controlling for orbit and asset class.
- **P2—Telemetry relevance.** Changes in propulsion, command capability, and propellant margin will predict provision revisions and disposal-method changes after controlling for asset age.
- **P3—Fallback materiality.** For high-cost active-removal scenarios, low-probability fallback paths will explain a disproportionate share of recognized liability and sensitivity.
- **P4—Network-obligation divergence.** The economic impairment unit will be broader than the obligation unit in integrated constellations, producing simultaneous network-level recoverability analysis and object-level provision accounting.
- **P5—Disclosure quality.** Dedicated orbital-obligation disclosures will be associated with lower estimation uncertainty and stronger reconciliation between decommissioning events, asset write-offs, and cash outflows.
- **P6—Regulatory shock.** Changes from 25-year to five-year disposal horizons will increase recognized liabilities primarily through timing, monitoring, and fallback-cost channels rather than through satellite construction cost.
- **P7—Bias detection.** Provision reductions unsupported by telemetry, legal release, or executed servicing evidence will cluster around financing and performance targets more than operational disposal outcomes.

12.3 Empirical data architecture

Future research can combine SEC and IFRS filings, FCC licensing databases, ESA and NASA mission documentation, space-object catalogs, reentry records, anomaly disclosures, insurance data, public

servicing procurements, launch prices, and operator sustainability reports. Object-level legal terms will remain the principal constraint. Researchers should avoid inferring an obligation solely from an orbital catalog or public disposal guideline.

12.4 Validation metrics

- Out-of-sample error in predicted settlement cost and timing.
- Calibration of self-disposal and active-removal probabilities by satellite generation and orbit.
- Timeliness of provision revisions after public anomalies or regulatory changes.
- Consistency between asset write-offs, disposal completion, and liability settlement.
- Comparability of recognized liability per obligated object after controlling for license and disposal method.
- Incremental explanatory power over simple rules based on satellite count, gross book value, or a fixed percentage of launch cost.

12.5 Pre-registration and falsification

Empirical tests should pre-register the obligation population, legal-scope criteria, settlement scenarios, cost-source hierarchy, telemetry variables, probability calibration, discount-rate method, materiality threshold, and rejection criteria. ORDOM is not supported if its additional complexity fails to improve prediction, reconciliation, or faithful representation relative to a simpler model. A fixed-price contractual deorbit obligation should outperform ORDOM's full scenario architecture because the simpler model matches the facts.

13. Boundary Conditions and Limitations

ORDOM is unnecessary when no present obligation exists, settlement cost is immaterial, or a fixed contractual amount and date require no complex model. It is also unnecessary when a satellite naturally decays and no future cash outflow, monitoring duty, or constructive commitment remains. The framework should not create provisions merely because the orbital environment would benefit from removal.

The model depends on legal analysis. International treaties often place responsibility at state level, while national law and contracts allocate cost to operators. Public licensing summaries may omit amendments, waivers, indemnities, or enforcement arrangements. An accounting model cannot repair incomplete legal evidence.

Active-removal markets are immature. Prices can be project-specific, capacity constrained, and dependent on launch availability, target geometry, capture interfaces, export controls, and liability allocation. Long-horizon cost estimates can therefore contain substantial model risk. ORDOM responds with evidence hierarchies and sensitivity, not false precision.

Telemetry can be incomplete, classified, or affected by survivorship bias. A failed satellite can stop transmitting precisely when the estimate becomes most important. Portfolio statistics can understate tail risk for unique objects. Engineering specialists must communicate uncertainty ranges rather than point estimates alone.

The distinction between ordinary operating costs and settlement expenditure remains judgmental. Ground operations can support both service and disposal. A shared mission-control center can be a corporate asset, an operating cost, and part of the provision's future cash flows. Allocation must be incremental, reasonable, and consistent with how the obligation will actually be settled.

The IAS 37 improvement project could change discount-rate and measurement requirements. ORDOM identifies current and proposed assumptions separately, but final amendments may require transition. US GAAP could also evolve as orbital obligations become more common, although the FASB decided in January 2026 not to add a project on ASC 410-20 to its technical agenda.

The public-company examples are incomplete by design. They demonstrate operating events and disclosed obligations, not the full accounting model. The paper does not infer that the absence of a separately labeled provision means no liability exists; amounts can be immaterial, aggregated, contractually transferred, or outside recognition criteria.

14. Conclusion

The end of an orbital asset's life creates an accounting problem that begins before reentry and can continue after the asset's economic value has disappeared. The physical object, the network that generates cash flows, and the legal or constructive duty to retire the object can exist at different units and on different timelines. Conventional accounting can fail when it treats planned disposal as automatically recognized, ignores a conditional legal duty because exercise is remote, writes off the asset and the liability together, or records the social cost of debris without a present obligation.

ORDOM translates orbital-retirement architecture into a disciplined IFRS and US GAAP process. It establishes a present-obligation gate; identifies the object, batch, license, or contract to which the duty attaches; constructs complete settlement scenarios; measures operations, passivation, monitoring, maneuver, servicing, and removal costs; updates probability and timing from controlled telemetry; and reconciles the liability to asset accounting, impairment, settlement, and disclosure.

The framework preserves the central differences between IAS 37 and ASC 410-20. IFRS can include constructive obligations and uses a best-estimate provision with current-rate remeasurement and IFRIC 1 asset adjustments. US GAAP focuses on legal AROs, fair value, credit-adjusted risk-free rate layers, and accretion. A common operational register can support both frameworks without blending their measurement objectives.

Public evidence shows the model's practical relevance. The FCC's five-year rule and disposal-plan disclosures connect licensing to timing and fuel planning. ESA's success thresholds and active-removal interfaces connect mission design to fallback scenarios. Iridium demonstrates authorization-specific deorbit terms and a historical probability-weighted satellite ARO. Spire demonstrates that decommissioning losses and liability measurement are distinct. Sirius XM demonstrates that retirement, execution, and completed deorbit can occur at different dates.

Taken together, MCDM, RLCAM, NUIM, and ORDOM form four connected layers of a broader space-accounting architecture:

$$\text{Consumption} \rightarrow \text{Cost allocation} \rightarrow \text{Recoverability} \rightarrow \text{Retirement} \tag{17}$$

The next layer should integrate the resulting accounting policies and metrics into a sector-specific orbital-asset and liability disclosure model. The strategic objective is not to create standards outside IFRS or US GAAP. It is to provide the operational discipline required to apply existing standards faithfully to the infrastructure, obligations, and stewardship challenges of the space economy.

Appendix A. Formula Use Map

Table A.1. Formula use map.

Eq.	Formula purpose	Primary use	Key control
(1)	Integrated model sequence	Positions ORDOM within the broader conceptual architecture without creating an implementation prerequisite.	ORDOM is fully autonomous; the sequence creates no implementation prerequisite.
(2)	Present-obligation gate	Tests past event, obligation source, and no realistic alternative.	Recognition before measurement.
(3)	IFRS recognition	Maps IAS 37 recognition conditions.	Separate legal/constructive scope and probability.
(4)	US GAAP recognition	Maps ASC 410-20 legal-obligation and estimability test.	Conditional timing does not remove legal duty.
(5)	Settlement cost stack	Builds scenario cash flows.	Incremental, directly attributable, no societal externalities.
(6)	Expected cost	Aggregates scenario probability and nominal cost.	Probabilities sum to one.
(7)	Self-disposal/fallback tree	Captures assisted disposal and active removal.	Fallback is recognized only within obligation scope and after the Section 5.9 ADR estimability gate.
(8)	Telemetry vector	Organizes disposal-readiness evidence.	Controlled data lineage.
(9)	Bayesian telemetry update	Converts priors and telemetry likelihoods into normalized posterior probabilities; the canonical equation is reproduced below.	Version-controlled priors, likelihoods, calibration cohorts, and overrides. Canonical Equation—Bayesian telemetry update (identical to Section 5.11): $$p(S_s \mid T_{i,t}) = \frac{p(T_{i,t} \mid S_s) \cdot p_0(S_s)}{\sum_{k=1}^{S} p\,(T_{i,t} \mid S_k) \cdot p_0(S_k)}$$
(10)	IFRS provision	Discounted best-estimate implementation.	No risk or inflation double-counting; IFRIC 1 increases are capped by IAS 36 recoverability, with any excess in profit or loss.
(11)	US GAAP ARO	Fair-value measurement architecture.	Market-participant and CARF-rate support; Level 3 estimates cannot bypass reasonable estimability.
(12)	Portfolio aggregation	Scales measurement to homogeneous fleets.	Object-level exceptions preserved.
(13)	Liability rollforward	Reconciles opening to closing balance.	Every movement linked to obligation register.
(14)	Disposal-success shortfall	Governance indicator against required threshold.	Not used directly as liability amount.
(15)	Sensitivity	Explains effect of a selected assumption change.	Reasonably possible, not cherry-picked.
(16)	Dual-reporting engine	Uses one fact base with separate IFRS and US GAAP models.	No framework blending.
(17)	Lifecycle conclusion	Summarizes integrated architecture.	Preserves the conceptual sequence and ORDOM's full modular autonomy.

Appendix B. Illustrative ORDOM Workbook Structure

Table B.1. Illustrative workbook modules.

Module	Required fields	Output/control
B.1 Object master	Object ID, launch, orbit, altitude, owner/operator, asset ID, generation, status.	Reconcile to asset register and external object catalog.
B.2 Obligation source	License, law, contract, public commitment, effective date, past event, legal owner.	IFRS and US GAAP scope flags.
B.3 Disposal architecture	Method, deadline, passivation, natural decay, graveyard orbit, removal interface.	Scenario-tree selection.
B.4 Telemetry snapshot	Fuel, propulsion, ADCS, TT&C, power, passivation, altitude, drag, conjunction, ADR readiness.	Time-stamped telemetry vector and likelihood inputs with approval and data-lineage controls.
B.5 Scenario probabilities	Scenario ID, baseline prior, likelihood function, posterior probability, calibration cohort, evidence, override reason.	Bayesian normalization, sum-to-one, zero-probability, versioning, and exception checks.
B.6 Cost model	Operations, maneuver, tracking, passivation, reentry, servicing, removal, regulatory.	Source hierarchy and currency/inflation controls.
B.7 Timing and rates	Settlement dates, maturity distribution, nominal/real basis, IFRS rate, US rate layer.	Framework-specific present value.
B.8 Accounting bridge	Provision/ARO, ARC, depreciation, impairment, settlement, reimbursement.	Journal-entry and general-ledger reconciliation.
B.9 Disclosure	Rollforward class, timing band, sensitivity, judgments, operational metrics.	Financial-statement note and sustainability cross-reference.
B.10 Back-test	Predicted and actual method, date, cost, settlement difference.	Bias analysis and model recalibration.

Appendix C. Illustrative Orbital Retirement and Debris Obligations Note

The following note is an illustrative disclosure template and is not based on the accounting records of any named company.

C.1 Accounting policy

The Group recognizes obligations for orbital retirement, decommissioning, and related debris-mitigation activities when it has a present legal or constructive obligation as a result of a past event, an outflow is probable, and the amount can be estimated reliably. Obligations primarily relate to post-mission disposal, passivation, tracking, regulatory reporting, and, for specified assets, contracted active removal. Provisions are measured at the present value of the best estimate of settlement expenditure. The corresponding initial amount is included in the cost of the related satellite or other long-lived asset when the asset-recognition criteria are met. Changes are accounted for in accordance with IFRIC 1. A US GAAP reporter would replace this policy with the applicable ASC 410-20 legal-obligation and fair-value policy.

C.2 Nature and scope

For illustration, assume that at year-end the Group operated 140 orbital objects subject to legally enforceable disposal requirements and 25 additional objects subject to a sufficiently specific constructive zero-debris commitment under IFRS. Obligations are measured by homogeneous cohorts based on orbit, satellite generation, disposal method, and license terms, with individual measurement for objects experiencing propulsion, command, or other anomalies. The principal settlement methods are controlled or uncontrolled atmospheric reentry, orbit lowering, graveyard-orbit transfer, and passivation. Active removal is included only for cohorts for which the Group bears an unavoidable fallback responsibility. These figures are hypothetical and are not derived from any named entity.

C.3 Carrying amount and movements

Table C.1. Illustrative liability rollforward.

US$ millions	Current year	Prior year
Opening balance	36.8	32.0
New obligations incurred	5.1	3.9
Revisions to cost and probability	7.9	2.1
Changes in discount rates	0.8	(0.4)
Unwinding/accretion	1.6	1.4
Settlements	(3.4)	(2.2)
Releases/transfers	(0.6)	—
Closing balance	48.2	36.8
Current portion	4.2	3.5
Noncurrent portion	44.0	33.3

C.4 Key assumptions and sensitivity

The most significant assumptions are the probability of successful self-disposal, the cost and availability of active-removal services, the timing of end-of-mission activities, inflation in launch and mission-operations costs, and the discount rate. In this hypothetical disclosure, a five-percentage-point decrease in self-disposal probability would increase the provision by approximately US$4.6 million; a one-year delay in expected settlement would

change the provision by approximately US$1.9 million; and a 50-basis-point decrease in the discount rate would increase the provision by approximately US$2.7 million. The sensitivity amounts are illustrative and not company data.

C.5 Unrecognized exposures

The Group also operates objects for which post-mission disposal is planned but no present legal or constructive obligation has been identified. The estimated operational expenditure for those activities is excluded from the provision. In addition, potential active-removal exposure for certain failed objects is disclosed separately when the applicable recognition criteria are not met or the amount cannot be estimated with sufficient reliability. The Group reassesses the legal basis, expected settlement method, and estimability as procurement and servicing markets develop. The precise wording must be adapted to the applicable IFRS or US GAAP reporting framework.

C.6 Connected operational metrics

Table C.2. Illustrative aggregated operational metrics.

Metric	Current year	Prior year
Objects entering disposal phase	12	9
Completed compliant disposals	10	7
Objects with unresolved disposal anomaly	2	1
Weighted-average expected settlement horizon	7.4 years	8.1 years
Objects in self-disposal success band above 90%	86%	89%
Objects with active-removal-compatible interface	72%	65%

Appendix D. Glossary of Acronyms and Variables

This glossary consolidates the principal accounting, regulatory, aerospace, technical, and mathematical terms used throughout ORDOM. Definitions are specific to the framework and should be read together with the recognition and measurement requirements discussed in the main text.

Table D.1. Glossary of acronyms and variables.

Category	Term or symbol	ORDOM definition
Accounting and regulatory	ADR	Active debris removal.
Accounting and regulatory	ARC	Asset retirement cost capitalized as part of the related long-lived asset.
Accounting and regulatory	ARO	Asset retirement obligation under US GAAP.
Accounting and regulatory	ASC	FASB Accounting Standards Codification.
Accounting and regulatory	CARF	Credit-adjusted risk-free rate used for US GAAP ARO liability layers.
Accounting and regulatory	CGU	Cash-generating unit under IAS 36.
Accounting and regulatory	COPUOS	United Nations Committee on the Peaceful Uses of Outer Space.
Accounting and regulatory	CSRD	Corporate Sustainability Reporting Directive of the European Union.
Accounting and regulatory	DSR	Design-Science Research.
Accounting and regulatory	ESG	Environmental, social, and governance.
Accounting and regulatory	FASB	Financial Accounting Standards Board.
Accounting and regulatory	FCC	Federal Communications Commission.
Accounting and regulatory	GAAP	Generally accepted accounting principles.
Accounting and regulatory	IAS	International Accounting Standard.
Accounting and regulatory	IASB	International Accounting Standards Board.
Accounting and regulatory	IFRIC	IFRS Interpretations Committee; also used in the titles of IFRIC Interpretations.
Accounting and regulatory	IFRS	IFRS Accounting Standards.
Accounting and regulatory	ISSB	International Sustainability Standards Board.
Accounting and regulatory	SEC	US Securities and Exchange Commission.
Aerospace and technical	ADCS	Attitude determination and control system.
Aerospace and technical	Conjunction	A predicted close approach between two orbiting objects that may require assessment or collision-avoidance action.
Aerospace and technical	GEO	Geostationary Earth orbit.
Aerospace and technical	LEO	Low-Earth orbit.
Aerospace and technical	Passivation	End-of-mission actions that reduce stored energy, including battery discharge, tank venting, pressure relief, and transmitter shutdown.
Aerospace and technical	TT&C	Telemetry, tracking, and command.
ORDOM notation	i	Individual spacecraft, upper stage, hosted payload, gateway, or other long-lived asset.
ORDOM notation	g	Homogeneous obligation group or launch batch.
ORDOM notation	s	Settlement scenario.
ORDOM notation	t	Reporting date.
ORDOM notation	$PE_{i,t}$	Past-event indicator for object i at reporting date t.
ORDOM notation	$OS_{i,t}$	Obligation-source indicator: legal, contractual, or constructive, as permitted by the applicable framework.
ORDOM notation	$OA_{i,t}$	No-realistic-alternative or unavoidable-obligation indicator.
ORDOM notation	$PO_{i,t}$	Present-obligation gate result.
ORDOM notation	$R^{Ifrs}{}_{i,t}$	Recognition decision under IFRS.
ORDOM notation	$R^{Us}{}_{i,t}$	Recognition decision under US GAAP.
ORDOM notation	$p_{i,s,t}$	Probability assigned to settlement scenario s.
ORDOM notation	$C_{i,s,t}$	Nominal settlement cash flow for object i under scenario s.

ORDOM notation	C_ADR	Active debris removal cost included only when a present obligation exists and the amount passes the ORDOM estimability gate.
ORDOM notation	$\tau_{i,s,t}$	Expected time to settlement under scenario s.
ORDOM notation	r^{Ifrs}_{t}	IFRS discount rate consistent with IAS 37 at reporting date t.
ORDOM notation	$r^{carf}_{i,l}$	US GAAP credit-adjusted risk-free rate assigned to liability layer l.
ORDOM notation	$P^{Ifrs}_{i,t}$	IFRS provision for orbital retirement and related obligations.
ORDOM notation	$ARO^{Us}_{i,t}$	US GAAP asset retirement obligation.
ORDOM notation	$ARC_{i,t}$	Associated capitalized asset retirement cost.
ORDOM notation	$T_{i,t}$	Telemetry and disposal-readiness vector.
ORDOM notation	$Q_{i,t}$	Probability of successful self-disposal.
ORDOM notation	$\Delta P_{i,t}$	Change in the recognized obligation during the reporting period.

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